MATH4511 Quantitative Methods for Fixed Income Securities

Lecture Note 2

Introduction to interest rate derivatives (Part 2)

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Objective of Chapter 2

In earlier chapter, we study some basic types of interest rate derivatives (bond, floating rate note and forward contract) and their application in portfolio management and risk management. We also explore the pricing approach of those derivatives using no arbitrage pricing principle.

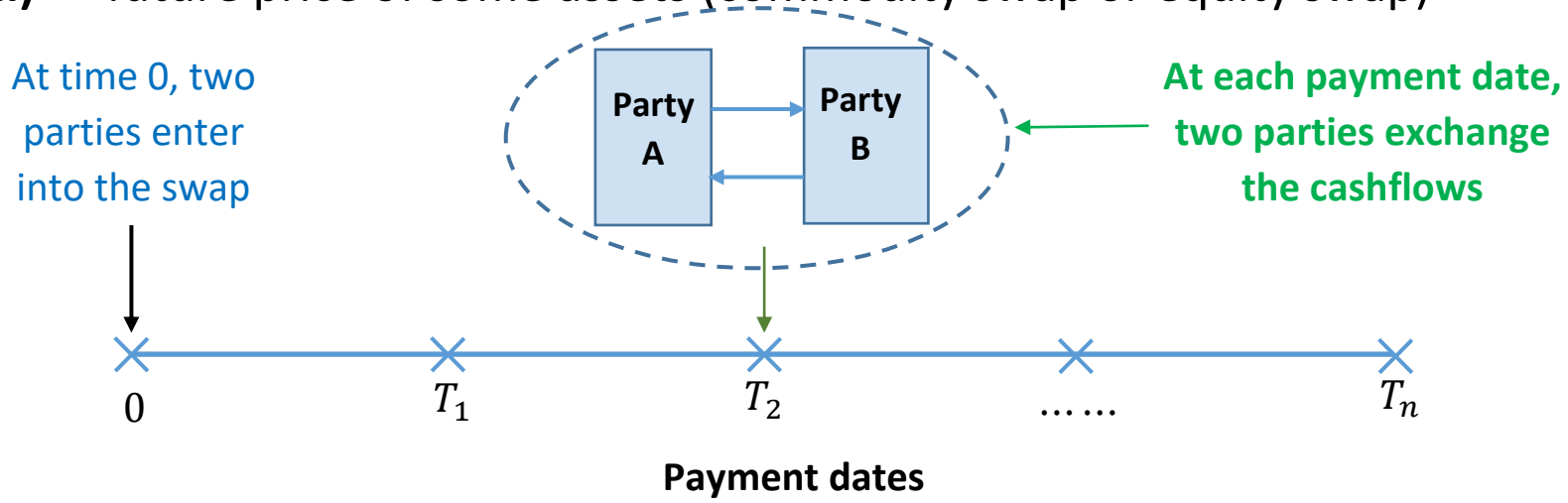
In this Chapter, we shall study two popular types of interest rate derivatives that are commonly traded in the financial market: Swap and options. We shall study the following:

- Swap – Interest rate swap, commodity swap, equity swap and foreign currency swap.
- Bond options, callable bond and puttable bond , interest rate cap, interest rate floor (options on interest rate), swaptions (options on swap)

Swap

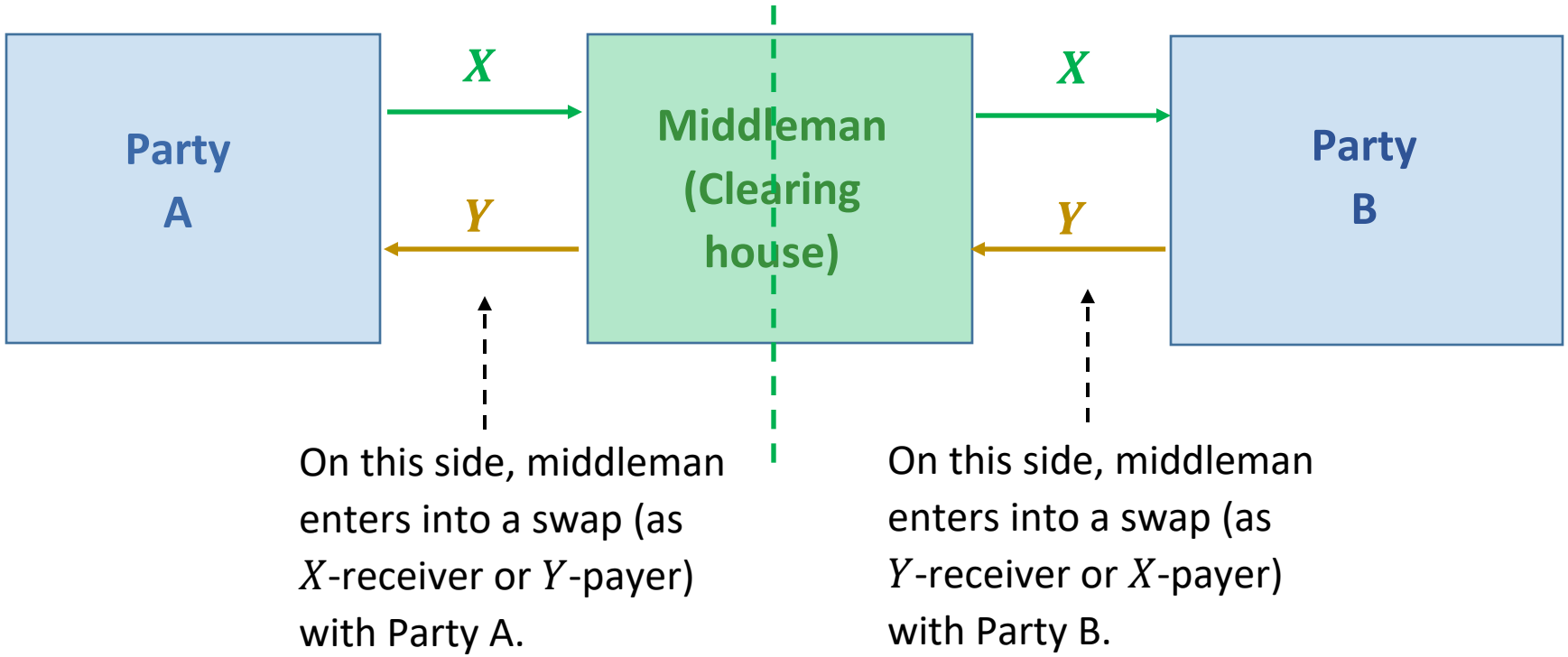
Swap is an agreement between two parties to exchange cash flows in future.

- The swap contract specifies the time (called *payment dates*) at which the payments are made from both parties.
- The amount of cash flows involved in each payment is calculated based on
  - **(i)** predetermined interest rate or market interest rate in future (interest rate swap),
  - **(ii)** interest rate/exchange rate of foreign currency (currency swap) or
  - **(iii)** future price of some assets (commodity swap or equity swap)



In practice, the transaction of swap is done through a financial institution. The financial institution serves as a *middleman* and process all payments involved in the swap.

Apparently, the financial institution has entered into *two* swaps with two parties respectively and the payments of two swap offset each other.



Role of the financial institution in swap contract

This middleman plays an important role in the whole transaction:

1. Minimize the potential default risk faced by two parties
   It is possible that one party may not fulfill the obligation if the loss is larger than his/her expectation and another party will suffer. When one of the parties default, the financial institution needs to fuifill the obligation with another party. Hence, both parties will be free from the loss due to potential default of another party.
   (*Note: Since all default risk is shifted to the financial institution, so it will demand a positive cost for compensation.)
2. Serve as a market makers
   In general, it is not easy to have two parties entering into a same swap at the same time. In this case, the financial institution will enter into a swap without having an offsetting swap with another party.
   (*Note: Without offsetting swap, the institution needs to offset its risk using other instruments such as bonds, forward contract (or FRA) etc.)

Interest rate swap

This is a swap which two parties exchange interest rate payments over a specified period of time. An interest rate swap can be described as follows:

- The amount of interest rate payment is calculated based on some fixed principal. It is known as *principal* (denoted it by N).
- One party is *fixed rate payer (or floating rate receiver)* who will pay fixed interest rate for floating interest rate (calculated by some market reference rate such as LIBOR). Another payer is *floating rate payer* (*or fixed rate receiver*) who will pay floating interest rate for fixed interest rate.
- The actual amount of interest rate payment (fixed or floating) involved at each payment date is defined as the total amount of interest earned by the principal N over the last payment period (calculated based on floating rate/ fixed rate). That is,

$$Payment\ amount = N \times R_{[T_{j-1},T_j]},$$

where $R_{[T_{j-1},T_j]}$ is the *effective interest rate* over the time interval $[T_{j-1}, T_j]$.

Example 1 (An example of interest rate swap)

We consider a 3-year interest rate swap which 6-month LIBOR is exchanged for a fixed interest rate at 3.8% every 6 months and the first exchange will be made after 6 months. Both interest rates are quoted as an *annual nominal interest rate payable semi-annually*.

Suppose that the notional principal is $N = \$100000$, then

- The amount of each fixed payment (known as fixed leg) is

$$Fixed\ leg\ payment = \$100000 \times \frac{0.038}{2} = \$1900.$$

- The amount of ith floating payment (known as floating leg), payable at time $T_i = \frac{i}{2}$, is given by

$$Floating\ leg\ payment = \$100000 \times \frac{L_{[T_{i-1},T_i]}}{2},$$

Here $L_{[T_{i-1},T_i]} = L_{\left[\frac{i-1}{2},\frac{i}{2}\right]}$ is the 6-month LIBOR rate applied over $\left[\frac{i-1}{2},\frac{i}{2}\right]$, quoted as annual nominal interest rate.

Suppose the *actual* 6-months LIBOR rates at different times are given by

	Now	0.5 years	1 year	1.5 years	2 years	2.5 years	3 years
6-months LIBOR $L_{[T_{i-1},T_i]}$	3.2%	3.6%	4.1%	4.8%	4.2%	3.5%	3.6%

Then the actual amount of 6 floating rate payments are determined to be

Payment	**1st**	**2nd**	**3rd**	**4th**	**5th**	**6th**
Floating rate leg	\$1600	\$1800	\$2050	\$2400	\$2100	\$1750

- The amount of kth floating rate payment is determined by the LIBOR rate determined at previous payment date.
- In practice, the LIBOR rates in future are not known currently. So we only know the amount of *first* floating rate payment and the amounts of other payments are not known (and are random).

Application of interest rate swap in risk management and portfolio management

1. *Use of swap in transforming a liability (e.g. loan)*

   We consider the following scenario: A company borrows an amount of N from a bank and run its business. The loan charges interest at a floating rate of $\{1\ year\ LIBOR\} + 1\%$ (quoted as annual effective interest rate) and the payment is made annually:

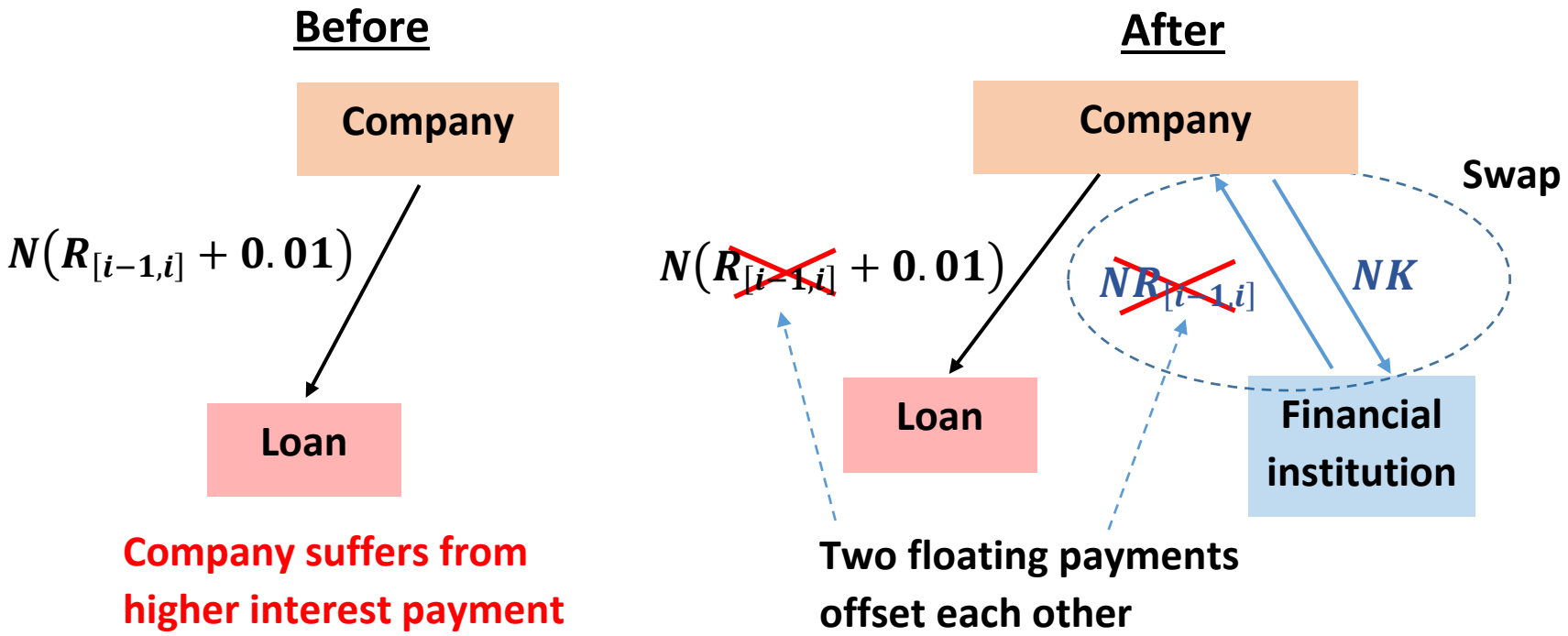
   - The company only repays the interest charged at intermediate repayment date and it repays the loan principal L at last repayment date.

   Suppose that the company foresees that the interest rate is likely to grow in future. This will increase the cost (interest payment) faced the company and this will cause an inefficient outcome.

   Question: Is it possible for the company to *control the cost*?

To resolve this problem, the company can enter into a swap and exchange fixed rate payment (with fixed annual interest rate $K_{[i-1,i]} = K$) for floating rate payment (calculated based on 1-year LIBOR) annually.

- By entering into swap, the floating rate payment received from the swap can offset the interest payment of the loan so that the net payment paid by the company at each payment date becomes fixed.



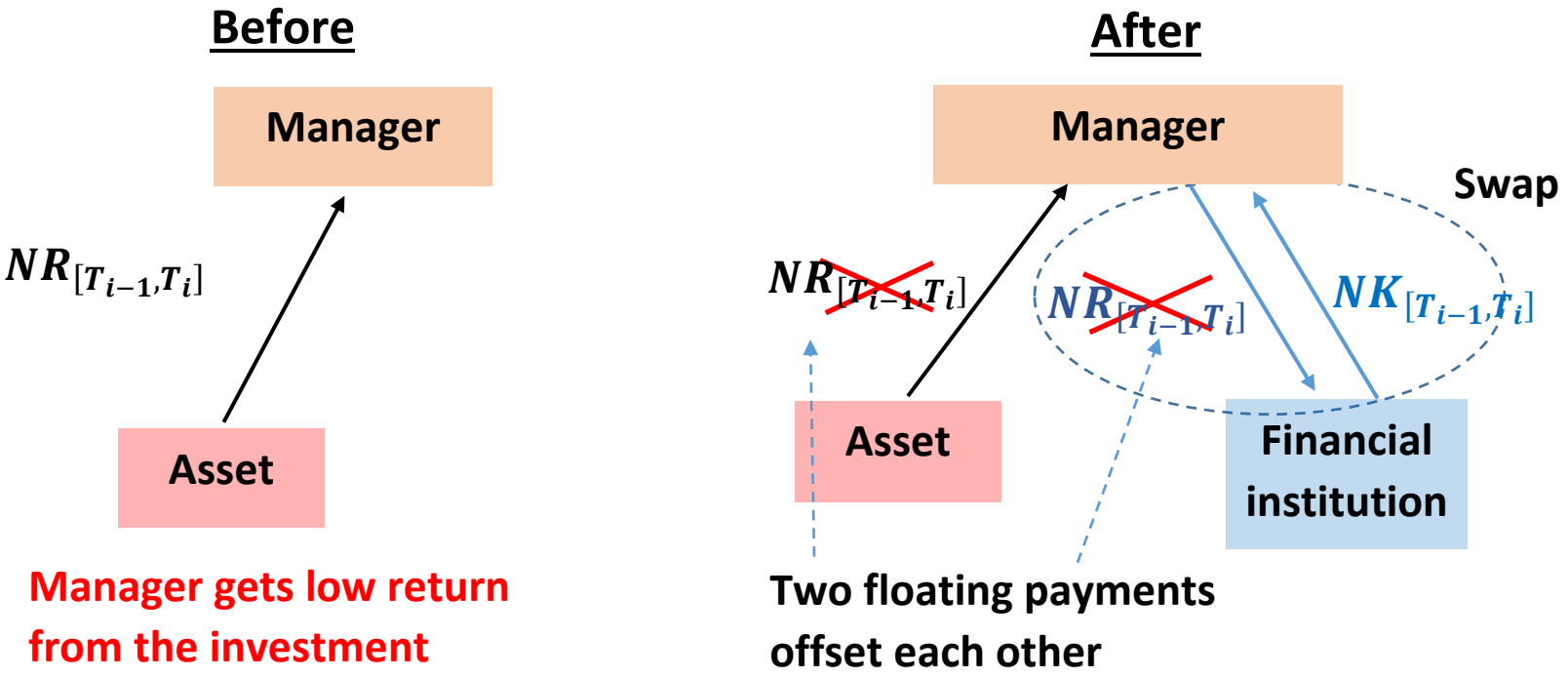
2. Use of swap in managing the portfolio

On the other hand, interest rate swap is also useful in "transform" the assets in the investment portfolio and allows investors to earn a higher return.

To see this, we consider the case when a fund manager is engaged in an investment which earns floating interest rate (e.g. investing in floating rate bond, lending the money at floating rate) periodically.

However, the manager forecasts from the market data that the market interest rate is likely to go down in future. This will decrease the manager's future return from the investment.

To avoid the potential low future return, the manager can enter into an interest rate swap (with notional principal N) which he/she exchanges the floating rate payment for fixed rate payment (with fixed rate $K_{[T_{i-1},T_i]}$, quoted as effective interest rate) with a financial institution periodically.



We observe that the manager is "holding an asset which generates a fixed return". In other words, the swap transforms the assets earning floating rate return into the assets earning fixed rate return.

Valuation of interest rate swap

Next, we shall examine the pricing issue of the interest rate swap and we would like to study the following issues:

1. Given that the swap is a mutual arrangement between two parties to exchange payments (similar to forward contract or FRA), what should be the fair value of fixed rate K (called *swap rate*)?
2. Suppose that a swap is initialized at time 0, what will be the value of the swap at future time t?

Model Setup

We consider the following interest rate swap with notional principal N:

- There are n payment dates and are denoted by T, $2T$, …, nT (i.e. we assume implicitly that the payments made regularly).
- The amount of ith fixed payment is $NK_{[(i-1)T,iT]} = NK$, where K is the effective fixed rate applied over the last payment period $[(i-1)T, iT]$.
- The amount of ith floating payment is $NR_{[(i-1)T,iT]}$, where $R_{[(i-1)T,iT]}$ is the effective market interest rate (e.g. LIBOR) applied over $[(i-1)T, iT]$.

Determination of swap rate K

Similar to the case for forward contract/FRA, one need to determine the value of swap at time 0. We consider the swap value at floating rate receiver side, Note that the swap consists of both floating rate payments (cash inflows) and fixed rate payments (cash outflows). Although the value of fixed rate payments can be computed using present value approach, one has to determine the value of floating rate payments using no arbitrage pricing principle.

Step 1: Value of fixed rate payment (fixed leg)

Using the standard discounting method, the present value of fixed rate payment (denoted by V_{fixed}) is given by

$$V_{fixed} = \sum_{i=1}^{n} (NK)B(0;iT) = NK \sum_{i=1}^{n} B(0;iT) \dots \dots (1).$$

Step 2: Value of floating rate payments (floating leg)

To find the value of ith floating rate payment, one needs to construct a trading strategy which can generate a payoff of $NR_{[(i-1)T,iT]}$ at time T_i.

We consider the following portfolio:

- Buy $(i-1)T$-years zero coupon bond with face value N;
- Short-sell iT-years zero coupon bond with same face value N;
- At time $(i-1)T$, the amount N received will be *reinvested* at interest rate $R_{[(i-1)T,iT]}$.

So the portfolio will generate a payoff of $N\left(1+R_{[(i-1)T,iT]}\right)-N=$ $NR_{[(i-1)T,iT]}$ at time T_i.

Using no arbitrage pricing principle, the current value of the ith floating rate payment (denoted by $V_{floating}^{(i)}$) equals the cost of constructing the portfolio:

$$V_{floating}^{(i)}=NB(0;(i-1)T)-NB(0;iT).$$

Hence, the total value of the floating leg is

$$V_{floating} = \sum_{i=1}^{n} \underbrace{[NB(0;(i-1)T) - NB(0:iT)]}_{V_{floating}^{(i)}} = N\left(\underbrace{B(0;0)}_{=1} - B(0:nT)\right).$$

Step 3: Find the value of the swap and determine the swap rate

The value of the swap (floating rate receiver position) at initialization date is

$$V_0 = V_{floating} - V_{fixed} = N\big(1 - B(0;nT)\big) - NK\sum_{i=1}^{n} B(0;iT).$$

To determine the swap rate, we note that swap is a mutual agreement between two parties to exchange payments. Since there is no side-payment involved, the swap value must be 0 at initialization date. So K must satisfy

$$\underbrace{N\big(1 - B(0;nT)\big) - NK\sum_{i=1}^{n} B(0;iT)}_{V_0} = 0 \Rightarrow K = \frac{1 - B(0;nT)}{\sum_{i=1}^{n} B(0;iT)}.$$

Determine the value of on-going swap

We consider an interest rate swap which is initialized at time 0. To compute the value of the swap at time t (where $t > 0$). One has to be careful that

- The swap value is calculated based on all *unpaid* payments made in future time (i.e. $T_i > t$).
- The amount of next floating payment is calculated based on the interest rate quoted at previous payment date, which has been fixed already.

We let T_k (where $k \leq n$) be the date of next payment. Then the value of the interest rate at time t can be computed as

$$V_t = \underbrace{NR_{[(k-1)T,kT]}B(t;kT) + \sum_{i=k+1}^{n}[NB(t;(i-1)T) - NB(t;iT)]}_{V_{floating}} - \underbrace{NK\sum_{i=1}^{n}B(t;iT)}_{V_{fixed}}$$

where $B(t;T)$ denotes the price of zero-coupon bond with face value 1 and maturity date T. (*Note: The summation will be vanished if $T_k = T_n$.)

Example 2 (Pricing of interest rate swap)

We consider a 3-year interest rate swap initialized recently which the payments are exchanged *annually*. You are given that

- The swap has notional principal $N = 100000$;
- The interest rate of floating leg payment is calculated by 1-year LIBOR rate.
- The interest rate of fixed leg payment is K, which is quoted as annual effective interest rate.
- The first exchange will be made 1 year after today.

The current term structure of interest rate is presented in the following table:

	6 months	1 year	1.5 year	2 years	2.5 years	3 years
LIBOR rate	1.5%	1.6%	1.8%	1.8%	2%	2.1%

Suppose that the term structure of interest rate after *6 months* is given by

	6 months	1 year	1.5 year	2 years	2.5 years	3 years
LIBOR rate	1.4%	1.7%	1.9%	2%	2%	2.3%

(*All rates are quoted as annual effective interest rate)

Question: Find the value of the swap (as floating rate receiver) at that time (i.e. after 6 months).

☺Solution

We first determine the swap rate K, which is determined at time 0. Note that the time 0 price of a T-year zero coupon bond (with face value 1) is given by

$$B(0;T) = e^{-L_{[0,T]}T}.$$

Hence, the swap rate K (as effective interest rate) is found to be

$$K \overset{T_i=i}{=} \frac{1-B(0;3)}{\underbrace{B(0;1)+B(0;2)+B(0;3)}_{\frac{1-B(0;T_n)}{\sum_{i=1}^{n} B(0;T_i)}}} = \frac{1-(1.021)^{-3}}{(1.016)^{-1}+(1.018)^{-2}+(1.021)^{-3}}$$
$$\approx 0.020924.$$

Next, we compute the swap value after 6 months (0.5 years). Note that the amount of next floating rate payment (to be paid at time 1) is $NR_{[0,1]} = 100000(0.016) = 1600$ (not $NR_{[0.5,1]}$).

Therefore, the present value of the fixed leg payment and floating leg payment are found to be

$$V_{floating} = \underbrace{1600B(0.5;1)}_{PV\ of\ 1st\ floating\ payment} + \underbrace{\sum_{i=2}^{3} 100000\big(B(0.5;i-1) - B(0.5;i)\big)}_{PV\ of\ 2nd\ \&\ 3rd\ floating\ payments}$$

$$= 1600(1.014)^{-0.5} + 100000((1.014)^{-0.5} - (1.019)^{-1.5})$$
$$+ 100000((1.019)^{-1.5} - (1.02)^{-2.5}) \approx 5726.291.$$

$$V_{fixed} = 100000K \sum_{i=1}^{3} B(0.5;i)$$
$$= 100000(0.020924)((1.014)^{-0.5} + (1.019)^{-1.5} + (1.02)^{-2.5})$$
$$= 6103.333.$$

Hence, the value of the swap (at floating rate receiver's side) is seen to be

$$V_{0.5} = V_{floating} - V_{fixed} = -377.042.$$

Example 3 (Determining spot rates from swap rates)

The following table shows the spot rates of different terms

	6 months	1 year	1.5 years	2 years
Spot rate	3%	3.2%	$s_0(1.5)$	$s_0(2)$

You are given that

- 1.5 years swap rate (where the payments are made semi-annually) is 3.3%
- 2 years swap rate (where the payments are made semi-annually) is 3.5%.

(*All spot rates and swap rates above are quoted as annual nominal interest rate compounded semi-annually.)

Determine 1.5-year spot rate and 2-year spot rate.

😊Solution

Recall that the swap rate is the fixed interest rate such that value of floating leg equals the value of fixed leg.

Since the 1.5-year swap rate is 3.3%, we have

$$\underbrace{\frac{0.033}{2}}_{K} = \frac{1 - B(0;1.5)}{\underbrace{\sum_{i=1}^{3} B(0;0.5i)}_{=\frac{1-B(0;nT)}{\sum_{i=1}^{n} B(0;iT)}}}$$

$$\Rightarrow 0.0165 = \frac{1 - \left(1 + \frac{s_0(1.5)}{2}\right)^{-3}}{\left(1 + \frac{0.03}{2}\right)^{-1} + \left(1 + \frac{0.032}{2}\right)^{-2} + \left(1 + \frac{s_0(1.5)}{2}\right)^{-3}} \Rightarrow s_0(1.5) = 0.033028.$$

On the other hand, we consider 2-year swap rate 3.5% and obtain

$$\underbrace{\frac{0.035}{2}}_{K} = \frac{1 - B(0;2)}{\underbrace{\sum_{i=1}^{4} B(0;0.5i)}_{=\frac{1-B(0;nT)}{\sum_{i=1}^{n} B(0;iT)}}}$$

$$\Rightarrow 0.0175 = \frac{1 - \left(1 + \frac{s_0(2)}{2}\right)^{-4}}{\left(1 + \frac{0.03}{2}\right)^{-1} + \left(1 + \frac{0.032}{2}\right)^{-2} + \left(1 + \frac{s_0(1.5)}{2}\right)^{-3} + \left(1 + \frac{s_0(2)}{2}\right)^{-4}}$$

$$\Rightarrow \cdots \Rightarrow s_0(2) = 0.035075.$$

Other equivalent pricing formula of interest rate swap

In this section, we explore other approaches in pricing interest rate swap. Recall that one key step in pricing swap is to construct a portfolio that replicates the floating leg payment $NR_{[(i-1)T,iT]}$ so that one can valuate the floating leg payments.

In fact, there are more than one possible portfolios that can replicate the same floating payment. By considering different portfolios, one can obtain different version of pricing formula.

1. *Pricing formula of swap using bond price*

   By adding a notional principal N to the cash inflow and cash outcome at the last payment date, the interest rate swap (floating rate receiver) can be viewed as a portfolio consists of

   - Buy one unit of floating rate note with face value N and coupon rate equals to the floating rate $R_{[(i-1)T,iT]}$;
   - Short-sell one unit of fixed rate coupon bond with face value N and coupon rate K (over a coupon period).

By no arbitrage pricing principle (or law of one price), the price of the interest rate swap (denoted by V_t) is given by

$$V_t = \begin{Bmatrix} Time\ t\ price\ of \\ floating\ rate\ note \end{Bmatrix} - \begin{Bmatrix} Time\ t\ price\ of \\ fixed\ rate\ bond \end{Bmatrix}.$$

This is another pricing formula for interest rate swap.

Remark: Another interpretation of swap rate K

Recall that the swap value is 0 at initialization date (i.e. $V_0 = 0$) and the price of the floating rate note at initialization date equals to face value N. So it follows that

$$\begin{Bmatrix} Price\ of\ fixed\ rate \\ bond\ at\ time\ 0 \end{Bmatrix} = N.$$

Hence, this formula provides another financial interpretation for the swap rate. In particular, swap rate is the coupon rate K in the fixed rate bond such that the bond is *sold at par* (i.e. the bond price equals its face value).

2. *Pricing formula of swap using forward rate*

Recall that the forward rate $K_{[(i-1)T,iT]}^{forward}$ reflects the "investor expectation" (in layman tone) to the future interest rate $R_{[(i-1)T,iT]}$, so it is possible to find the value of floating leg payment using forward rate.

To see this, we consider an on-going swap at time t (which the next payment will be made at kT ($> T$). We consider a floating rate payment payable at iT (with amount $NR_{[(i-1)T,iT]}$) where $i > k$.

To replicate this payment, we consider the following portfolio (constructed at <u>time t</u>):

- Enter into FRA over the period $[(i-1)T, iT]$ in long position with forward rate $K_{[(i-1)T,iT]}^{forward}$ and notional principal N. Then the investor will get an amount $NR_{[(i-1)T,iT]} - NK_{[(i-1)T,iT]}^{forward}$ at time iT).
- Buy zero coupon bond with face value $NK_{[(i-1)T,iT]}^{forward}$ and maturity date iT (This will generate a cash inflow $NK_{[(i-1)T,iT]}^{forward}$ at time iT.)

So the net payment of the portfolio at time iT is

$$\left(NR_{[(i-1)T,iT]} - NK^{forward}_{[(i-1)T,iT]}\right) + NK^{forward}_{[(i-1)T,iT]} = NR_{[(i-1)T,iT]}.$$

By no arbitrage pricing principle, the value of the floating leg payment equals to the current value of the portfolio. Since the value of FRA is 0, we have

$$\begin{Bmatrix} Present\ value\ of\ i^{th} \\ floating\ leg\ payment \end{Bmatrix} = NK^{forward}_{[(i-1)T,iT]} B(t; iT),$$

where $B(t; iT)$ denotes the current price of zero coupon bond with face value 1 and maturity date t.

Hence, the current value of the swap at time t can be expressed as

$$V_t = \underbrace{NR_{[(k-1)T,kT]} B(t, kT)}_{kth\ floating\ payment\ (fixed)} + \sum_{i=k+1}^{n} \underbrace{NK^{forward}_{[(i-1)T,iT]} B(t; iT)}_{\substack{ith\ floating\ payment \\ (expected)}}$$

$$- \sum_{i=k}^{n} NKB(t; iT).$$

Note that the payment $NK^{forward}_{[(i-1)T,iT]}$ can represent the market's (or investors) expectation on the amount of floating rate payment $NR_{[(i-1),iT]}$. The above formula suggests that one can compute the present value of floating leg payment by finding the "expected value" of the $NR_{[(i-1),iT]}$ and discount it back to time t. That is,

$$\left\{ \begin{matrix} Present\ value\ of\ i^{th} \\ floating\ leg\ payment \end{matrix} \right\} = \underbrace{NK^{forward}_{[(i-1)T,iT]}}_{\begin{matrix}\text{expected value} \\ of\ NR_{[(i-1),iT]}\end{matrix}} \times \underbrace{B(t;iT)}_{\begin{matrix}discounting \\ factor\end{matrix}}.$$

Remark:

In the above formula, the forward rate $K^{forward}_{[(i-1)T,iT]}$ should be calculated based on the spot rate quoted at time t since we assume that the investor enters into FRA at time t. That is,

$$K^{forward}_{[(i-1)T,iT]} = \frac{B(t;(i-1)T)}{B(t;iT)} - 1 = \frac{1+R_{[t,iT]}}{1+R_{[t,(i-1)T]}} - 1.$$

3. *Pricing formula of the swap using swap rate quoted at time t*

We consider an interest rate swap with swap rate K (quoted as effective interest rate over payment period) initialized at time 0.

Recall that the swap rate is the fixed rate applied to fixed rate payment such that *the value of interest rate swap is zero at the initialization date*.

We let K_t be the swap rate of the interest rate swap initialized at time t and the cashflows (after time t) are identical to that of the original interest rate swap.

Since the swap value equal to difference of present value of floating leg payment and present value of fixed leg payment, one can use the definition of swap rate (denoted by K_t) and deduce that

$$\begin{Bmatrix} Present\ value\ of \\ floating\ leg\ payments \\ at\ time\ t \end{Bmatrix} = \begin{Bmatrix} Present\ value\ of\ fixed\ leg \\ payments\ with\ coupon\ rate\ K_t \\ quoted\ at\ time\ t \end{Bmatrix}.$$

Hence, the value of the on-going interest rate swap at time t (denoted by V_t) is

$$V_t = \begin{Bmatrix} PV\ of\ floating \\ leg\ payments \end{Bmatrix} - \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K \end{Bmatrix}$$

$$= \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K_t \end{Bmatrix} - \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K \end{Bmatrix}$$

$$= \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K_t - K \end{Bmatrix}.$$

Example 4 (Pricing of swap using forward rate)

We consider an on-going swap which the floating rate receiver is agreed to pay fixed payment at a fixed rate of 8% (quoted as annual nominal interest rate compounded semi-annually) and receive a floating payment at 6-month LIBOR on a notional principal of $N = \$100000$ *every 6 months*. You are given that

- The swap has remaining life of 15 months.
- The current LIBOR rates for 3-month, 9-month and 15-month maturities, which are quoted as annual effective interest rate, are 9%, 9.5% and 10% respectively.
- The 6-month LIBOR rates at the previous payment date was 9.3%.

Calculate the value of this swap using forward rate approach.

😊Solution

We take $t = 0$ be the current time. To calculate the present value of floating rate payments, one needs to compute the forward rates applied over the period $[0.25, 0.75]$ (i.e. $K_{[0.25,0.75]}$)and $[0.75, 1.25]$ (i.e. $K_{[0.75,1.25]}$) .

The forward rates $K_{[0.25,0.75]}$ and $K_{[0.75,1.25]}$ are found to be

$$K_{[0.25,0.75]} = \frac{(1+0.095)^{0.75}}{(1+0.09)^{0.25}} - 1 \approx 0.04762,$$

$$K_{[0.75,1.25]} = \frac{(1+0.1)^{1.25}}{(1+0.095)^{0.75}} - 1 \approx 0.052399.$$

On the other hand, the amount of first floating rate payment and each fixed payment are $NR_{[-0.25,0.25]} = 100000(1.093^{0.5} - 1) = 4546.64$ and $NK = 100000\left(\frac{0.08}{2}\right) = 4000$ respectively.

Hence, the current value of the swap can be computed as

$$V_t = \underbrace{\frac{4546.64}{(1+0.09)^{0.25}} + \frac{0.04762(100000)}{(1+0.095)^{0.75}} + \frac{0.052399(100000)}{(1+0.1)^{1.25}}}_{Present\ value\ (Floating\ leg\ payment)}$$
$$\underbrace{- \left(\frac{4000}{(1+0.09)^{0.25}} + \frac{4000}{(1+0.095)^{0.75}} + \frac{4000}{(1+0.1)^{1.25}}\right)}_{Present\ value\ (Fixed\ leg\ payment)} \approx 2347.49.$$

Example 5 (Application of swap rate)

One year ago, an investor entered into a 4-year interest rate swap with annual payments and swap rate 6% (quoted as annual effective interest rate) as *floating rate receiver*.

The current swap rate of various maturities (with annual payments) are given as follows: (*All rates are quoted as annual effective interest rate)

	1 year	2 year	3 years	4 years
Swap rate	5.7%	5.9%	6.2%	6.5%

(a) What can you say about the profitability of the swap?

(b) Suppose that the interest drops and the current swap rates become 5.4% (1 year), 5.6% (2 years), 5.8% (3 years) and 6.1% (4 years) respectively. What can you say about the profitability of the swap?

If the swap becomes unprofitable, is it possible for the investor to stop the loss?

😊 Solution

(a) Recall that the value of on-going swap can be expressed as

$$V_t = \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K_t - K \end{Bmatrix},$$

where $K = 0.06$.

Since the swap has remaining life 3 years, we have $K_t = 0.062$ and $K_t - K > 0$. This implies that $V_t > 0$ and the swap is profitable.

(b) Then $K_t = 0.058$ and $K_t - K < 0$, so we have $V_t < 0$ and the swap is not profitable. To stop any future loss, the investor can enter into another 3-year swap as *fixed rate receiver*. Then the floating payments of those two swaps offset each other so that the net future payments become deterministic. The present value of the future payments (profit & loss of the investor) is

$$V_t = \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ coupon\ rate\ K_t - K \end{Bmatrix} = \sum_{i=1}^{3} N(K_t - K)B(0;i).$$

(Question: Can we calculate V_t?)

Other variants of interest rate swap

Example 6 (Deferred interest rate swap and its valuation)

We consider the following scenario. A company has made a 10-years floating rate loan. The loan charges interest at 1-year LIBOR rate (currently 1%) and the repayments are made annually. We assume that the company pays interest-due only in first 9 repayments and the loan principal, together with the interest-due, will be repaid at last repayment date (10th repayment).

Suppose that the company expects that the interest rate stays low during first 4 years and starts to climb after 4 years. To hedge the potential risk, the company may consider to enter into an interest rate swap and exchange the fixed rate payment for floating rate payment.

However, it may be too early to do so since the company may need to pay more during the first 4 years. To resolve this problem, the company can choose to enter into *deferred interest rate swap* which the first exchange is made 5 years after today (i.e. the first exchange is deferred by 4 years) and the exchanges are paid annually until year 10.

The term structure of interest rate at current time is given as follows:

	1 year	2 years	3 years	4 years	5 years	6 years	7 years	⋯	10 years
LIBOR rate	1%	1.2%	1.2%	1.2%	1.4%	1.4%	1.65%	…	1.65%

(*All rates are quoted as *annual effective interest rate*.)

Question: Determine the swap rate K (quoted as annual effective interest rate) for the deferred swap.

☺Solution

We first need to determine the current value of the deferred swap. Note that

- The present value of the fixed rate payment is

$$V_{fixed} = \sum_{i=5}^{10} \underbrace{NK \times B(0;i)}_{PV\ of\ i^{th}\ payment} = NK \sum_{i=5}^{10} \frac{1}{\left(1 + L_{[0,i]}\right)^{i}} = 5.333941NK.$$

- The present value of the floating rate payment is

$$V_{floating} = \sum_{i=5}^{10} \underbrace{NB(0;i-1) - NB(0;i)}_{PV\ of\ i^{th}\ payment} = N[B(0;4) - B(0;10)]$$
$$= 0.10437N.$$

Thus, the current value of the swap is

$$V_0 = V_{floating} - V_{fixed} = 0.10437N - 5.333941NK.$$

By setting $V_0 = 0$, we get $K = \frac{0.10437}{5.333941} \approx 0.019567$.

Remark of Example 6

- Deferred swap is also known as *forward swap* in financial community. It is because the deferred swap can be seen as a forward contract on swap which the investor is obligated to enter into the underlying swap at a fixed swap rate K at future time.

- Although the investor can choose entering into an ordinary interest rate swap at time 4 (instead of entering into deferred swap), he/she may need to pay higher fixed rate K since the swap rate K is calculated based on the interest rate quoted at time 4.

  For example, if the market interest rate at time 4 is given as follows:

	1 year	2 years	3 years	4 years	5 years	6 years
LIBOR rate	2%	2%	2.3%	2.3%	2.3%	2.6%

  Then the swap rate applied to the 7-year interest rate swap at that time is

$$K = \frac{1 - B(4;10)}{\sum_{i=1}^{6} B(4;4+i)}$$

$$= \frac{1 - (1.026)^{-6}}{(1.02)^{-1} + (1.02)^{-2} + (1.023)^{-3} + \cdots + (1.026)^{-6}} \approx 0.025771.$$

  which is higher than the swap rate in the deferred swap (i.e. 0.019567).

- One may ask whether it is more profitable for investor to enter into the 10-year swap now because of lower swap rate.
  By direct computation, the swap rate of 10-year interest rate swap is

$$K^* = \frac{1 - B(0;10)}{\sum_{i=1}^{10} B(0;i)} = \frac{0.150964}{9.218719} \approx 0.016376,$$

  which is lower than the swap rate ($K = 0.019567$) of the deferred swap. This is expected since the swap reflects the "*average value*" of the 1-year interest rate applied over the 10 years.

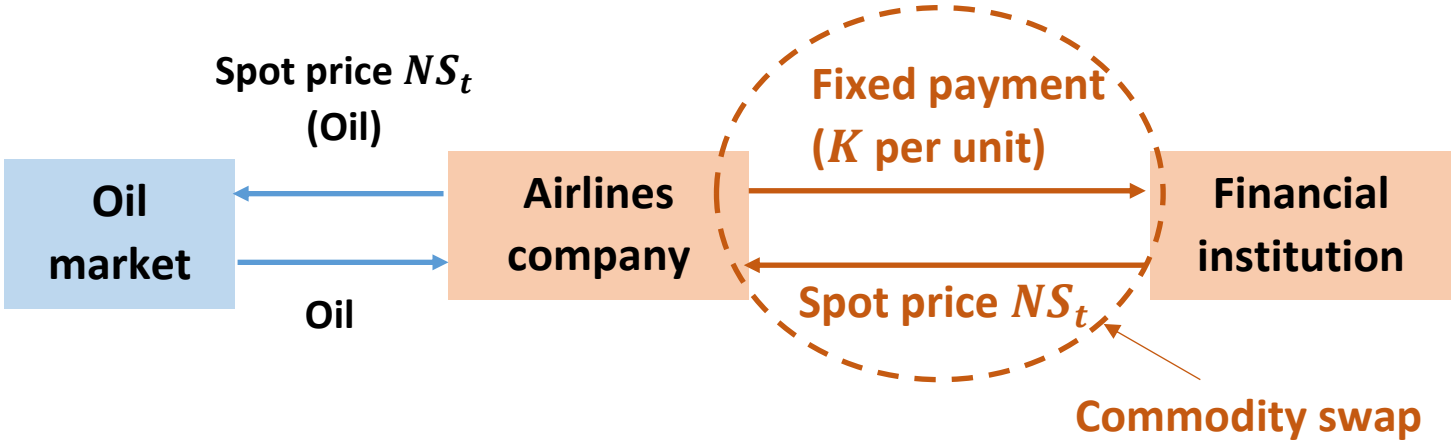
  Although the investor can enjoy a lower swap rate by entering into the 10-year swap now, he/she may need to make a larger payment in the first 4 payments (comparing with deferred swap). Suppose that the 1-year interest rate in the first 4 years remains to be around 1%, the investor needs to pay an additional $(0.016376 - 0.01)N = 0.006376N$ in the first 4 payments and this may override the benefit of paying less in the last 6 payments.

Commodity swap

We consider an airlines company that has to purchase oil regularly for running its operations. However, the price of oil fluctuates over time and the company suffers from risk due to the price fluctuations. It is important for the company to "control" the cost to secure its profit.

To hedge the market risk, the company may wish to enter into a *commodity swap* with a financial institution. Under the terms of the commodity swap,

- the airline company receives spot price for a certain units of oil at each exchange date.
- At the same time, the company pays a fixed amount K per unit.



Hence, the airline company can purchase the oil at a fixed and predetermined price ($\$K$ per unit). This allows the company to have a better control to its budget.

In this section, we would like to examine the following issues:

- How to determine the fair value of K (swap rate for commodity swap)?
- How to price an on-going commodity swap?

We consider a commodity swap which the payments are made at times $T, 2T, \ldots, nT$. To determine the swap rate, we first calculate the swap value (at floating rate receiver side) at the initialization date (time 0). Note that

- The present value of all fixed leg payments is

$$V_{fixed} = \sum_{i=1}^{n} NKB(0; iT) \quad or \quad \sum_{i=1}^{n} \frac{NK}{\left(1 + s_0(iT)\right)^{iT}}.$$

- Next, we consider the ith floating payment (with amount NS_{iT}) paid at time iT. To find the "present value" of this payment. We consider the following portfolio:

- ✓ Enter into iT-year forward contract on N units of the asset with forward price $f_{[0,iT]}$ *per unit*.
- ✓ Buying a iT-year zero coupon bond with face value $Nf_{[0,iT]}$.

Then the net payoff of this portfolio at time iT is $Nf_{[0,iT]} + N\left(S_t - f_{[0,iT]}\right) = NS_{iT}$, which is same as the amount of ith floating payment.

By no arbitrage pricing principle, the present value of ith floating payment equal to the value of the replicating portfolio:

$$V^i_{floating} = \underbrace{0}_{contract} + \underbrace{Nf_{[0,iT]}B(0;iT)}_{Bond} = Nf_{[0,iT]}B(0;iT).$$

Hence, the value of the swap at time 0 is given by

$$V_0 = \underbrace{\sum_{i=1}^{n} Nf_{[0,iT]}B(0;iT)}_{V_{floating}} - \underbrace{\sum_{i=1}^{n} NKB(0;iT)}_{V_{fixed}}.$$

By setting $V_0 = 0$, we deduce that the swap rate is given by

$$K = \frac{\sum_{i=1}^{n} N f_{[0,iT]} B(0;iT)}{\sum_{i=1}^{n} NB(0;iT)} = \sum_{i=1}^{n} \underbrace{\left(\frac{B(0;iT)}{\sum_{i=1}^{n} B(0;iT)} \right)}_{weight} f_{[0,iT]},$$

which is the weighted average of the forward prices of different maturities.

Computing the value of on-going swap at time t ($t > 0$).

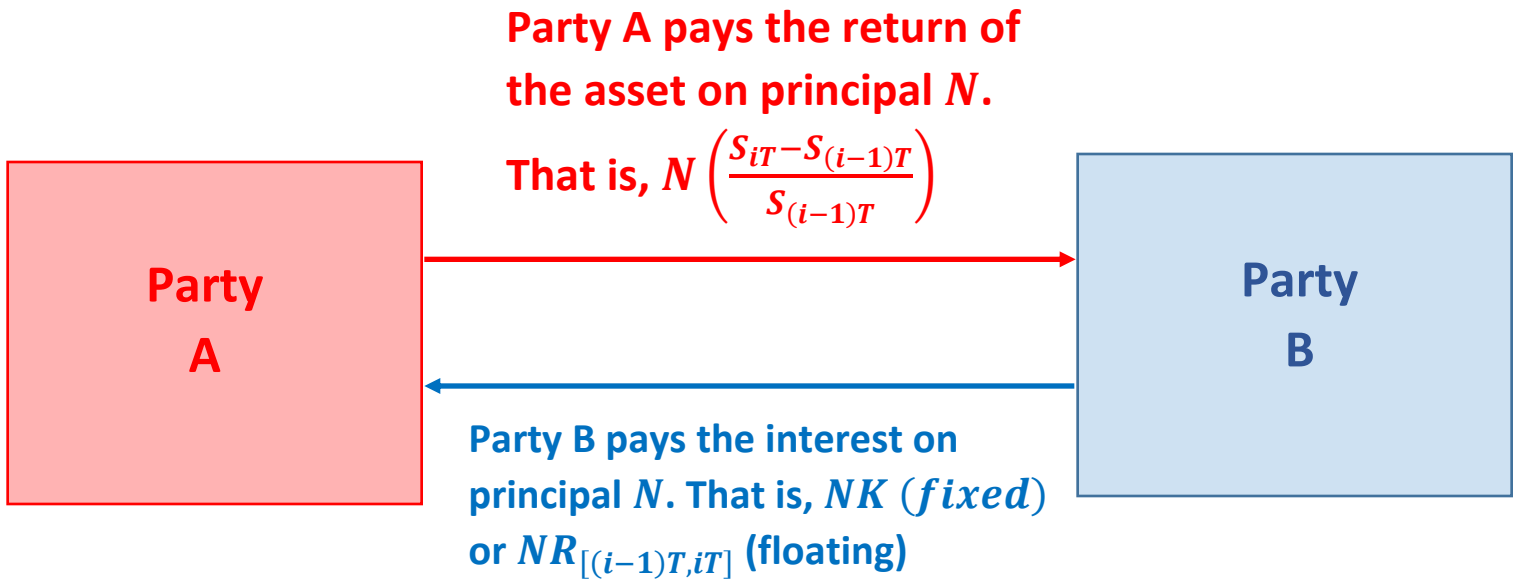
We let kT be the next payment (exchange) date (i.e. $(k-1)T \leq t < kT$), one can use the above result and show that the current value of the swap is

$$V_t = \underbrace{\sum_{i=k}^{n} N f_{[t,iT]} B(t;iT)}_{V_{floating}} - \underbrace{\sum_{i=k}^{n} NKB(t;iT)}_{V_{fixed}} = N \sum_{i=k}^{n} (K_t - K) B(t;iT),$$

where K_t denotes the swap rate of the commodity swap quoted at time t.

Equity swap

In an equity swap, one party agrees to pay the return on an asset (e.g. stock, index) on a notional principal N over a series of scheduled payment date, while the other promises to pay a fixed interest or floating interest (based on market interest rate) on the same notional principal N. The actual transaction in the equity swap at a ith payment date can be summarized by the following figure:



(*Note: S_t denotes the value/price of the asset at time t.)

Why do we need equity swap?

- It allows the investor to earn from the return of a target asset/index without holding the asset. This is useful when the investor has difficulty in buying the asset physically (e.g. the asset is foreign stock/fund or equity index, transaction cost and tax).
- Suppose that an individual predicts the performance of a stock is going worse in future and he/she would like to earn profit from this, the investor can use the equity swap to secure the profit without shortselling the stock (which may not be allowed in some market).
- Suppose that an individual owns a large amount of a company stock. If the individual knows that the stock performance is going to be bad in future, equity swap allows the individual to hedge against the downside loss *without selling the stock* to the market. This can protect him the benefit from holding the stock such as dividend or voting right.

Valuation of equity swap

Next, we shall derive the pricing formula of an equity swap using no arbitrage pricing principle. Similar to the case for interest rate swap, we compute

1. The value of the swap at the initialization date. This allows us to determine the swap rate of the swap if necessary.
2. The value of the on-going swap at future time t

To start with, we first consider the equity swap which the floating interest rate payment $NR_{[(i-1)T,iT]}$ is exchanged for equity return $N\left(\frac{E_{iT}-E_{(i-1)T}}{E_{(i-1)T}}\right)$.

1. <u>Value of equity swap at initialization date (time 0)</u>
   We shall compute the present values of floating rate leg and equity leg respectively. Using the result derived from interest rate swap, the present value of floating leg is given by

$$V_{floating} = \sum_{i=1}^{n} [NB(0;(i-1)T) - NB(0;iT)] = N - NB(0;nT).$$

Next, we consider the ith equity leg payment. To find its value, we consider the following portfolio:

- Buy a $(i-1)T$-year bond with face value N. At time $(i-1)T$, invest the amount N received into the asset and
- shortsell a T-year bond with face value N.

The portfolio will generate a cashflow $\frac{N}{S_{(i-1)T}} S_{iT} - N = N\left(\frac{S_{iT}-S_{(i-1)T}}{S_{(i-1)T}}\right)$ at time iT which matches the ith equity payment.

By no arbitrage pricing principle, the current value of ith equity leg payment equals to the value of the portfolio. That is,

$$V^i_{equity} = NB_{[0,(i-1)T]} - NB_{[0,iT]}.$$

Thus, the present value of the equity leg payments is

$$V_{equity} = \sum_{i=1}^{n} \underbrace{\left(NB(0;(i-1)T) - NB(0;iT)\right)}_{V^i_{equity}} = NB(0;0) - NB(0;nT)$$

$$= N - NB(0;nT).$$

Therefore, the current value of the swap at time 0 is

$$V_0 = \underbrace{[N - NB(0;nT)]}_{V_{equity}} - \underbrace{[N - NB(0;nT)]}_{V_{floating}} = 0.$$

It appears that no side payment is needed for two parties to enter into the contract.

2. Value of on-going equity swap (at intermediate time t)

Suppose that $T_{k-1} \leq t < T_k$, one can compute the value of two payments as follows:

- The present value of the floating leg payments is

$$\begin{aligned} V_{floating} &= NR_{[(k-1)T,kT]}B(t;kT) + \sum_{i=k+1}^{n} [NB(t;(i-1)T) - NB(t;iT)] \\ &= NR_{[(k-1)T,kT]}B(t;kT) + NB(t,kT) - NB(t,nT). \end{aligned}$$

- To get the present value of the equity leg payments, we note that
  - ✓ For $i \geq k+1$, the present value of ith equity payment is

$$V_{equity}^{i} = NB_{[0,(i-1)T]} - NB_{[0,iT]}.$$

✓ To find the value of kth equity payment paid at time kT, we consider the following portfolio: Buy $\frac{N}{S_{(k-1)T}}$ units of the equity and shortsell a zero coupon bond maturing at time kT with face value N, then the portfolio value at time kT is $\frac{N}{S_{(k-1)T}} S_{kT} - N = N\left(\frac{S_{kT}-S_{(k-1)T}}{S_{(k-1)T}}\right)$, which is same as the amount of kth equity payment.

By no arbitrage pricing principle, the value of kth equity payment equals to the value of the replicating portfolio. That is,

$$V_{equity}^{k} = \frac{N}{S_{(k-1)T}} S_t - NB(t; kT).$$

Combining, the present value of the equity leg payment is

$$V_{equity} = \underbrace{\frac{N}{S_{(k-1)T}} S_t - NB(t; kT)}_{V_{floating}^{k}} + \sum_{i=k+1}^{n} \underbrace{\left(NB_{[0,(i-1)T]} - NB_{[0,iT]}\right)}_{V_{floating}^{i},\ i \geq k+1}$$

$$= \frac{N}{S_{(k-1)T}} S_t - NB(t; nT).$$

- Hence, the value of the on-going swap is found to be

$$V_t = \underbrace{\left[\frac{N}{S_{(k-1)T}} S_t - NB(t;nT)\right]}_{V_{equity}} - \underbrace{\left[NR_{[(k-1)T,kT]} B(t;kT) + NB(t,kT) - NB(t,nT)\right]}_{V_{floating}}$$

$$= \frac{N}{S_{(k-1)T}} S_t - N\left(1 + R_{[(k-1)T,kT]}\right) B(t;KT).$$

On the other hand, one can consider the equity swap which fixed rate payment (with rate K) is exchanged for equity payment. Using similar method (left as exercise), one can show that

- The swap rate K at time 0 (such that the swap value is 0) is

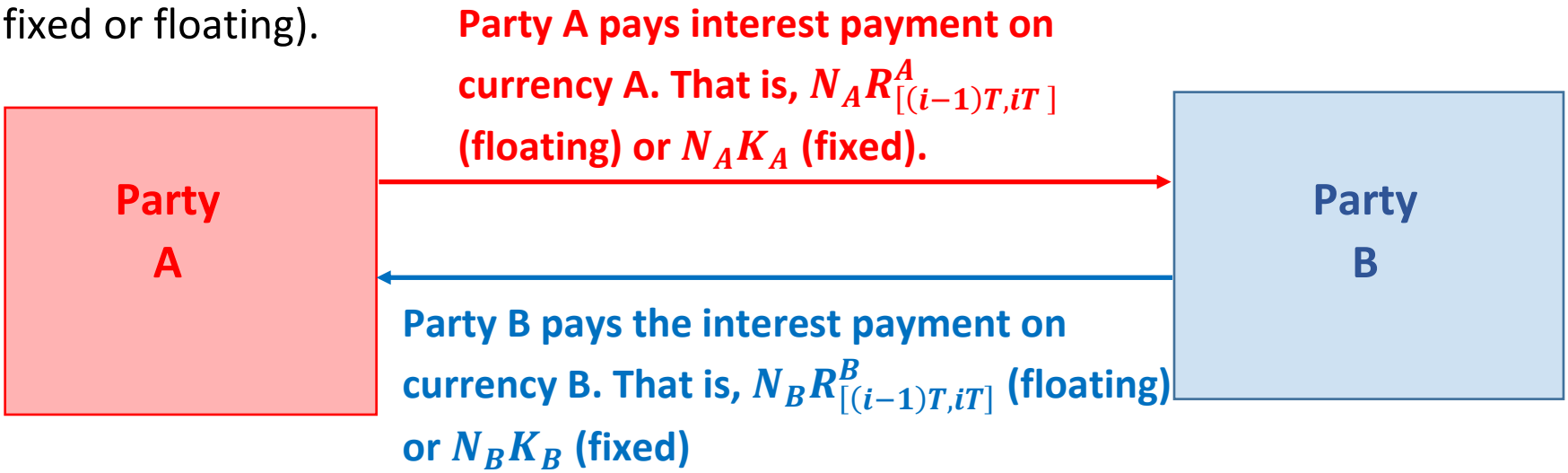
$$K = \frac{1 - B(0;nT)}{\sum_{i=1}^{n} B(0;iT)}.$$

- The swap value at time t (where $(k-1)T \leq t < kT$) is

$$V_t = \frac{N}{S_{(k-1)T}} S_t - NB(t;nT) - \sum_{i=k}^{n} NKB(t;kT).$$

Foreign currency Swap

Current swap allows the holder to exchange interest payment in one currency for interest payment in another currency for a period of time. The amount of each payment depends on a fixed notional principal and a certain interest rate (either fixed or floating).



*__Note__: Usually, the notional principals N_A and N_B are determined such that they have the same values at the initialization date. That is, $N_B = S_B N_A$, where S_B denoted the price (in currency A) of one unit of currency B.

*__Note 2:__ Different from interest rate swap, *notional principals are also exchanged* at the initialization date and end date of the currency swap.

Application of currency swap

To motivate the use of currency swap, we consider the following scenario:

There are two companies (Company A and Company B) in different countries:

- Company A, which is a US company, would like to launch its business in Japan. To do so, it needs to raise capital in Japanese yen.
- Company B, which is a Japanese company, would like to expand its business in US. It needs capital in US dollars.

Although both companies can borrow the required capital from their local bank, they need to pay higher borrowing cost since the bank may charge a higher interest rate for borrowing foreign currency. The table below shows borrowing rates of two companies:

	Borrowing Cost	
	USD	JPN
Company A (US)	4.5%	**4.2%**
Company B (Japan)	**5.2%**	3%

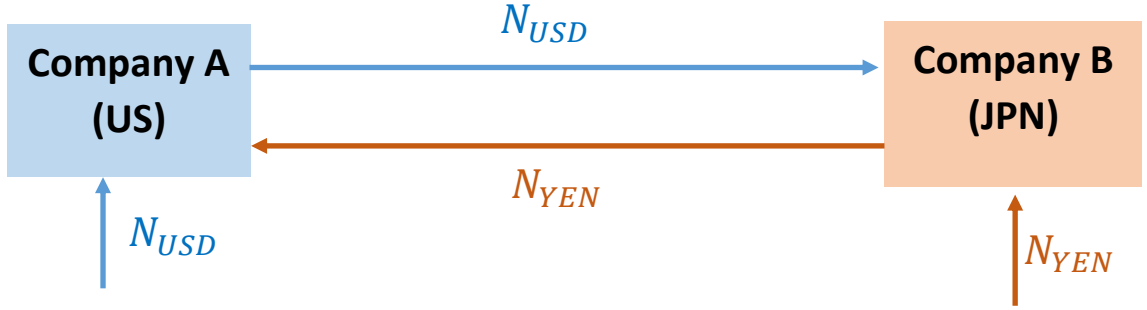
(*Note: The above rates are quoted as annual effective interest rate. The repayments are made annually.)

We observe that two companies need to pay a higher interest rate payment if they borrow the foreign currency directly from its local financial institution (comparing with another countries).

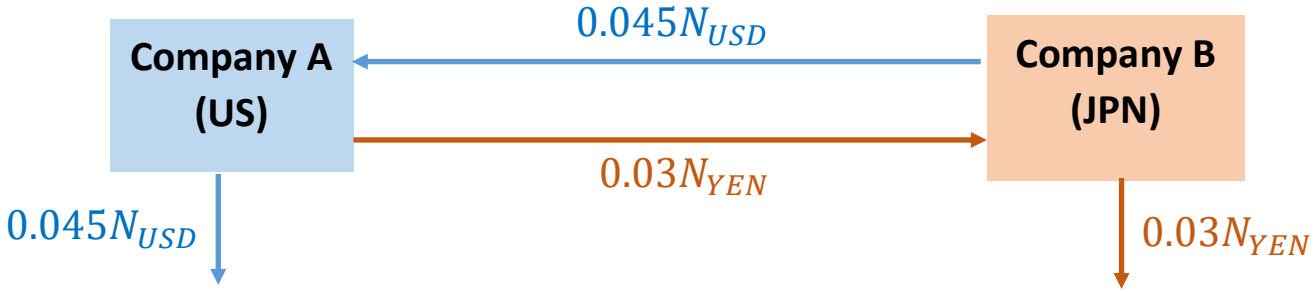
Question: Is it possible for them to reduce the cost by some strategies?

Note that both companies can enjoy a lower cost if they borrow their local currency. In other words, they have *comparative advantage* in their local currency market. So it is possible for two companies to reduce the cost if they "co-operate" and *borrow the capital for another company*. To do so, they can enter into a foreign currency swap:

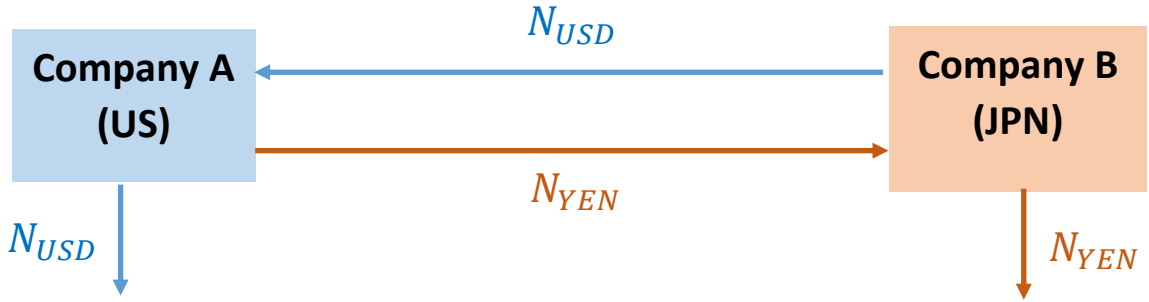
Step 1: (At time 0) Company A borrows USD dollar at 4.5% and Company B borrows JPN dollar at 3%. They enter into currency swap and exchange the notional principals.



Step 2: At i^{th} repayment date (i.e. time T_i), two companies exchange the interest rate payment through the swap. Company A pays the interest payment of company B's loan and company B pays the interest payment of company A's loan.

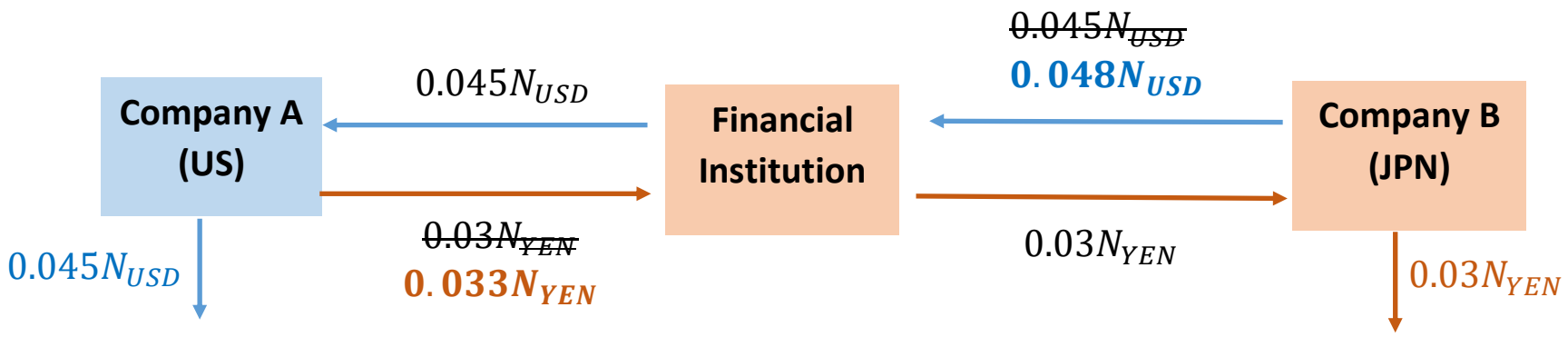


Step 3: (At time T_n) At the last repayment date, two companies exchange the notional principal again and each company use the principal received to repay its outstanding loan completely.



Using currency swap, we see that company A can now borrow Japanese Yen at 3% (instead of 4.2%) and company B can now borrow USD dollar at 4.5% instead of 5.2%. This greatly reduces their borrowing cost.

In general, the two companies may not know each other. In this case, they can enter currency swap through a financial institution (cleaning house). The institution will handle all payments involved in the currency swap between two parties. It will charge some cost to each of two parties (due to credit risk, service fee):



- Under the currency swap, the financial institution earns $0.003N_{YEN}$ (in JPN currency) and $0.003N_{USD}$ (in USD currency)
- Although there will be a transaction cost, two companies are still willing to enter the swap and enjoy the lower borrowing cost.

Valuation of currency swap

Next, we examine the pricing of the currency swap. One challenge in the valuation process is that multiple currencies are involved in the swap and one has to convert all payments into same currency to complete the valuation process.

To start with, we consider the following currency swap which the payments in currency A is exchanged for payments in currency B.

(*To facilitate the analysis, We take currency A be the *domestic currency* and B be the *foreign currency*. The value of the currency swap is calculated in currency A.)

- There are n exchange dates in the swap. Namely, $T_1, T_2, \dots, T_n$, where $T_i = iT$.
- The notional principals of two currencies are N_A (currency A) and N_B (currency B) respectively. N_A, N_B are chosen such that they have the same value at time 0. That is, $N_A = S_{B,0}N_B$, where $S_{B,0}$ is the exchange rate at $t = 0$ (i.e. 1 unit of currency B $= \$S_{B,0}$ (in currency)).
- The interest rate applied in both currencies are fixed. We assume that the interest rates over *each payment period* are K_A (currency A) and K_B (currency B) respectively. (i.e. The swap is a fixed-fixed swap).

We consider the position which the investor receives payments in currency B (foreign currency receiver). To find the value of the currency swap, we need to construct a portfolio which generates the cashflows that are identical to that of the currency swap. Since all payments are fixed, one can consider the following portfolio (constructed at time 0):

- For each $i = 1,2, \dots, n$, we *buy* T_i-year foreign zero-coupon bond with face value $K_B N_B$ (in currency B) so that we can receive $K_B N_B$ at time T_i.
- For each $i = 1,2, \dots, n$, we *short-sell* T_i-year zero-coupon bond with face value $K_A N_A$ (in currency A) and repay $K_A N_A$ at time T_i. This generates a negative cashflow $N_A K_A$ at time T_i.
- (For last payments). We buy T_n-year foreign zero-coupon bond with face value N_B (in currency B) and short-sell T_n-year zero-coupon bond with face value N_A (in currency A).

Since the future payoffs of the portfolio match that of the currency swap, it follows from no arbitrage pricing principal that the value of the currency swap at time 0 equals the value of the portfolio.

We let V_0 be the price of currency swap, we have

$$V_0 = \underbrace{S_{B,0}}_{\substack{exchange \\ rate}} \underbrace{\left[\sum_{i=1}^{n} K_B N_B B_F(0;T_i) + N_B B_F(0;T_n)\right]}_{value\ (in\ currency\ B)} - \underbrace{\left[\sum_{i=1}^{n} K_A N_A B(0;T_i) + N_A B(0;T_n)\right]}_{value\ (in\ currency\ A)},$$

where $B_F(0,T)$ denotes the current price of foreign zero coupon bond with face value 1 (in currency B/ foreign currency).

Some remark about the pricing formula of currency swap

- *Pricing of on-going currency swap at time t*
  We let T_k be the next payment date of the currency swap (i.e. $T_{k-1} \leq t < T_k$).
  Using similar method, one can deduce that the price of the swap at time t is
  $$V_t = S_{B,t}\left[\sum_{i=k}^{n} K_B N_B B_F(t;T_i) + N_B B_F(t;T_n)\right] - \left[\sum_{i=k}^{n} K_A N_A B(0;T_i) + N_A B(0;T_n)\right].$$
- *Another pricing formula using forward price*
  On the other hand, one can derive the same formula using forward price of the foreign currency. This can be done by treating currency B as an asset and modifying the replicating portfolio.

To start with, we let $f_{[0,T]}$ be the T-year forward price for 1 unit of currency B. Recall from Chapter 1 that the forward price can be expressed as

$$f_{[0,T]} = \frac{S_{B,0} B_F(0;T)}{B(0;T)}.$$

We consider the following modified portfolio:

- ✓ For each $i = 1,2,\dots,n$, we *enter into T_i-year forward contract on $N_B K_B$ units of currency B* and *buy T_i-year zero coupon bond with face value $N_B K_B f_{[0,T]}$*. By doing so, we can get $N_B K_B$ unit of currency B at time T_i.
- ✓ For each $i = 1,2,\dots,n$, we *short-sell T_i*-year zero-coupon bond with face value $K_A N_A$ (in currency A) and repay $K_A N_A$ at time T_i. This generates a negative cashflow $N_A K_A$ at time T_i.
- ✓ (For last payments). We *enter into T_n-year forward contract on N_B units of currency B* and *buy T_n-year zero coupon bond with face value $N_B f_{[0,T]}$*. and short-sell T_n-year zero-coupon bond with face value N_A (in currency A).

One can check that this portfolio generates cashflows which are identical to the currency swap.

By no arbitrage pricing principle and recall the face that the contract value is 0 at initialization date 0, the current price of the swap can be expressed as

$$V_0 = \sum_{i=1}^{n} \Big(0 + K_B N_B f_{[0,T_i]} B(0;T_i)\Big) + \Big(0 + N_B f_{[0,T_n]} B(0;T_n)\Big) - \left[\sum_{i=k}^{n} K_A N_A B(0;T_i) + N_A B(0;T_n)\right]$$

Using the fact that $f_{[0,T]} = \frac{S_{B,0} B_F(0;T)}{B(0;T)}$, we can deduce that

$$V_0 = \sum_{i=1}^{n} \left(K_B N_B \underbrace{\left(\frac{S_{B,0} B_F(0;T_i)}{B(0;T_i)}\right)}_{f_{[0,T_i]}} B(0;T_i) \right) + \left(0 + N_B \underbrace{\left(\frac{S_{B,0} B_F(0;T_n)}{B(0;T_n)}\right)}_{f_{[0,T_n]}} B(0;T_n) \right) - \left[\sum_{i=k}^{n} K_A N_A B(0;T_i) + N_A B(0;T_n)\right]$$

$$= S_{B,0}\left[\sum_{i=1}^{n} K_B N_B B_F(0;T_i) + N_B B_F(0;T_n)\right] - \left[\sum_{i=k}^{n} K_A N_A B(0;T_i) + N_A B(0;T_n)\right].$$

Example 7 (Designing a currency swap)

There are two companies based in US and HK respectively. Suppose that company A (based in US) wishes to borrow Hong Kong dollar (of amount N_{HKD}) for 5 years and company B (based in Hong Kong) wishes to borrow US dollars (of amount N_{USD}) for 5 years.

The following table shows the borrowing rates quoted by their local bank.

	USD	HKD
Company A	3.0%	**5.6%**
Company B	**4.6%**	6.0%

(*Note 1: The borrowing rates are quoted as annual nominal interest rate compounded semi-annually.)
(*Note 2: Two companies will make interest payment semi-annually.)

(*Note 3: Here, company A is assumed to have better credit rating so that it can borrow the money at lower rate.)

The current exchange rate is $1USD = 7.8HKD$. The notional principals are taken to be $N_{USD} = USD\$10000$ and $N_{HKD} = \$78000$. So two principals have same values.

Question: Is it possible for the financial institution to design a 5-year currency swap (with semiannual payments) which all parties are equally better off? (That is, two companies can pay less interest payment and the institution can earn some profit.)

☺Solution

The design of the currency swap consists of two key steps:

Step 1: Identify a comparative advantage

Suppose that two companies are willing to borrow the money for another company and exchange the payments through the swap.

- ✓ Suppose that the company A borrows the USD dollar (with principal USD10000) for company B. At each repayment date, the company B can save up to $\$(7.8) \times \left[10000\left(\frac{0.046}{2} - \frac{0.03}{2}\right)\right] = 78000(0.008)$ (in HKD currency)
- ✓ Suppose that the company B borrows the HKD dollar (with principal HKD$78000). At each repayment date, the company A can save up to $\$78000 \times \left[\frac{0.056}{2} - \frac{0.06}{2}\right] = 78000(-0.002)$ (in HKD currency)

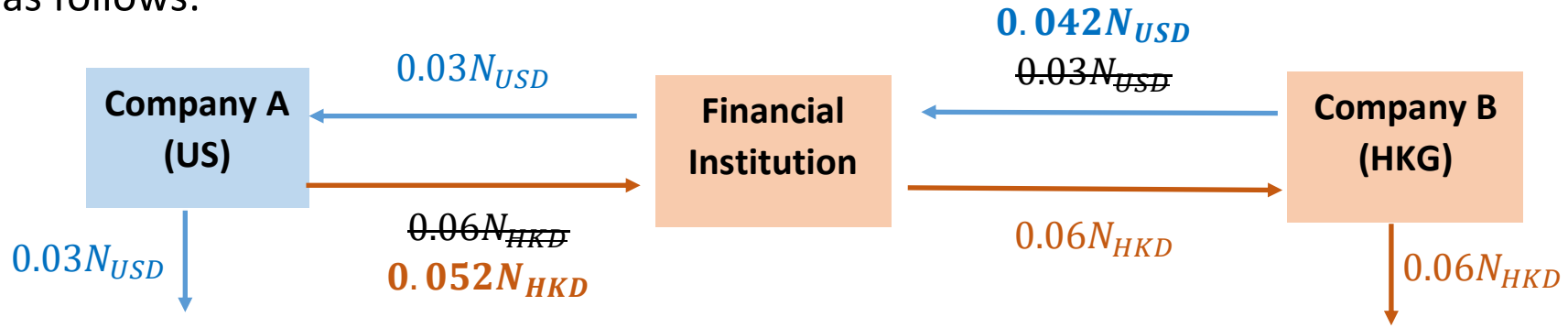
Then the total net profit is $78000(0.008 - 0.002) = 78000(0.006) > 0$ and there is an advantage if they cooperate.

Step 2: Design a currency swap

The idea is to distribute this additional benefit (of amount $78000(0.006)$) to three parties (two companies and financial institution) evenly so that they are equally better off. We start with the following currency swap:



It is clear that company A is not better off in this scenario since it has to pay more and the financial institution earns nothing. Thus, one needs to modify the cashflow as follows:



Under this swap, we have

- The company A can save up to $\$\left(\frac{0.056}{2}-\frac{0.052}{2}\right)N_{HKD}=\$(78000)(0.002)$.
- The company B can save $USD\left(\frac{0.046}{2}-\frac{0.042}{2}\right)N_{USD}=USD(10000)(0.002)$. Using the current exchange rate, the value is $\$(78000)(0.002)$.
- Using the current exchange rate, the net profit of the financial institution at each repayment date can be estimated as

$$Profit=\left(\left(\frac{0.042}{2}-\frac{0.03}{2}\right)N_{USD}\right)(7.8)-\left(\frac{0.052}{2}-\frac{0.06}{2}\right)N_{HKD}$$
$$=78000(0.002).$$

So they are equally better off (based on the data available today).

Remark of Example 9

Although two companies are strictly better off by entering currency swap, the financial institution is facing foreign currency risk since the actual profit depends on the exchange rate at the payment date (Here, we estimate the profit using today's exchange rate). To resolve this problem, the institution can enter into forward contract on USD currency so that it can convert USD into HKD at a fixed rate.

Example 8 (Valuation of foreign currency swap)

We consider the currency swap developed in previous example. Suppose that the term structure of interest rate of two currencies are both flat. You are given that

- The annual effective interest rate of USD dollar is 2.5% and
- The annual effective interest rate of HKD dollar is 4%.

Calculate the value of two currency swap (in HKD currency).

😊Solution

We consider the currency swap between company A and financial institution. Using the formula established earlier with $B(0;T)=\frac{1}{(1.04)^T}$ and $B_F(0;T)=\frac{1}{(1.025)^T}$, the value of the currency swap (on company A's side) is

$$V_0^A = 7.8\underbrace{\left(\sum_{i=1}^{10}\frac{10000\left(\frac{0.03}{2}\right)}{(1.025)^{\frac{i}{2}}}+\frac{10000}{(1.025)^5}\right)}_{value\ of\ USD\ payments}-\underbrace{\left(\sum_{i=1}^{10}\frac{78000\left(\frac{0.052}{2}\right)}{(1.04)^{\frac{i}{2}}}+\frac{78000}{(1.04)^5}\right)}_{vaue\ of\ HKD\ payments}$$
$$= 7.8(10240.95) - 82345.7 \approx -2466.3.$$

Similarly, the value of the currency swap between company B and the institution can be computed as

$$V_0^B = \underbrace{\left(\sum_{i=1}^{10} \frac{78000\left(\frac{0.06}{2}\right)}{(1.04)^{\frac{i}{2}}} + \frac{78000}{(1.04)^5}\right)}_{vaue\ of\ HKD\ payments} - 7.8\underbrace{\left(\sum_{i=1}^{10} \frac{10000\left(\frac{0.042}{2}\right)}{(1.025)^{\frac{i}{2}}} + \frac{10000}{(1.025)^5}\right)}_{value\ of\ USD\ payments}$$

$$= 85151.15 - 7.8(10801.91) = 896.252.$$

Remark of Example 8 (Calculating the value of the currency swap)

The above example demonstrates an easy approach in calculating the value of fixed-fixed currency swap.

- For each currency leg (HKD or USD), we calculate the present value of all unpaid cashflows using the interest rate of that currency.
- If the currency is foreign currency, convert the value into domestic currency using the current foreign exchange rate.

Example 9 (Fixed-to-floating currency swap and its valuation)

Suppose that company A wishes to borrow US dollars (of amount N_{USD}) and company B wishes to borrow HKD dollars for 5 years (of amount N_{HKD}). The following table shows the borrowing rate offered to two companies:

	HK dollars	US dollars
Company A	3.0%	LIBOR + 1%
Company B	4.5%	LIBOR + 1.5%

(*Note: All rates are quoted as annual effective interest rate. In addition, we assume that the company will make interest payment to the loan annually.)

You are given that

- The current exchange rate is $1USD = \$7.8\ HKD$.
- The term structure of both currencies are flat. The annual effective interest rates of HK dollar and US dollar are 2.5% and 3.5% respectively.
- The notional principals are chosen such that $N_{HKD} = 7.8N_{USD}$ (two principals have same values at initialization date).

(a) Is it possible to design a currency swap for company A and company B such that the financial institution can earn a spread of 50 basis points ($= 0.5\%$) and the swap is equally attractive to both companies?

(b) Hence, determine the price of the currency swap (at company A's side) between company A and the financial institution.

☺ Solution

(a) To design a suitable swap, we first calculate the additional benefit earned at each repayment date if two companies enter into the currency swap and borrow the capital for another company.

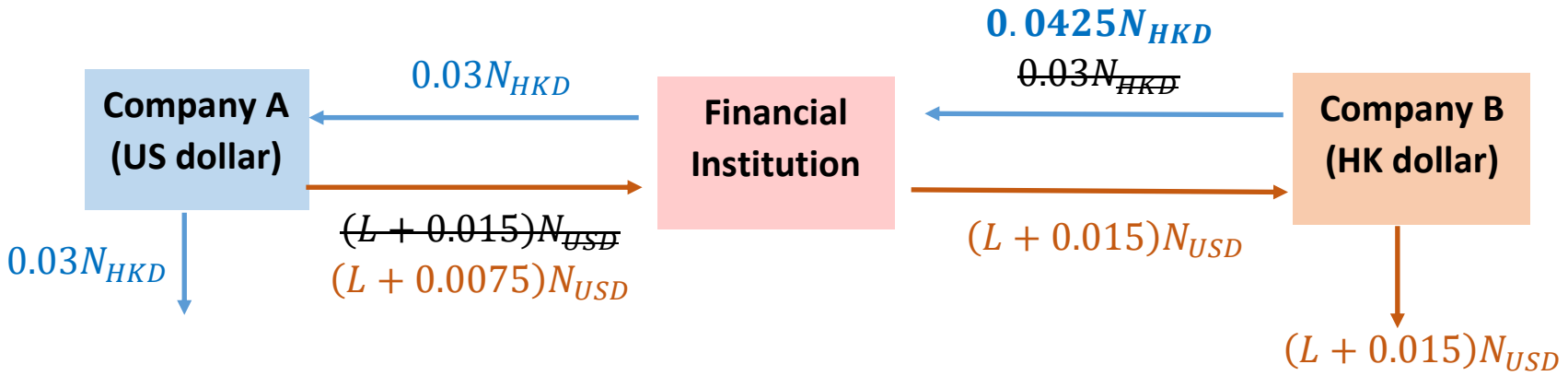
- If company A borrows HK dollars for company B, then company B can save an amount $\$N_{HKD}(0.045 - 0.03) = 0.015N_{HKD}$.
- If company B borrows US dollars for company A, then company A will "lose" an amount $\$7.8\underbrace{N_{USD}\big(L + 0.015 - (L + 0.01)\big)}_{in\ USD\ currency} = N_{HKD}(0.005)$.

So the net profit is $0.015N_{HKD} - N_{HKD}(0.005) = 0.01N_{HKD}$.

Since the financial institution (FI) needs to earn a profit of $0.005N_{HKD} = 0.005(7.8)N_{USD}$ from the swap, so each company should earn

$$\frac{1}{2}\left(\underbrace{0.01N_{HKD}}_{Total\ profit} - \underbrace{0.005N_{HKD}}_{profit\ (F.I.)}\right) = 0.0025N_{HKD}$$ (i.e. 0.25% of N_{HKD}.

Given this fact, one can construct the currency swap as follows:



As a verification, one can check that the profit made by financial institution at each payment date is

$$Profit = (0.0425 - 0.03)N_{HKD} - 7.8(L + 0.015 - L - 0.0075)N_{USD}$$
$$= 0.005N_{HKD}.$$

(b) To calculate the value of the currency swap, we calculate the present value of the cash inflows (in HKD currency) and the cash outflows (in USD currency).

- Since spot rates of HKD currency is $L_{[0,T]} = 2.5\%$ for all T, hence the present value of cash inflows is

$$V_{HKD} = \sum_{i=1}^{5} \frac{0.03N_{HKD}}{(1.025)^i} + \frac{N_{HKD}}{(1.025)^i} = 1.023229N_{HKD}.$$

- Using similar pricing approach as in valuating floating leg in interest rate swap, the present value of the cash outflows (in HKD currency) is

$$V_{USD} = 7.8\left(\sum_{i=1}^{5} N_{USD}\left(\underbrace{\frac{1}{(1.035)^{i-1}}}_{B(0;i-1)} - \underbrace{\frac{1}{(1.035)^{i}}}_{B(0;i)}\right) + \sum_{i=1}^{5} \frac{0.0075N_{USD}}{(1.035)^i} + \frac{N_{USD}}{(1.035)^i}\right) = 1.033863N_{HKD}.$$

Thus, the current price of the currency swap (in HKD currency) is

$$V_0 = \underbrace{1.023229N_{HKD}}_{V_{HKD}} - \underbrace{1.033863N_{HKD}}_{V_{USD}} = -0.01063N_{HKD}.$$

Options

Roughly speaking, options is a financial contract which gives the holder (in long position) a right, but not obligation, to buy/sell an asset at a specific price X (known as strike price) on a specific date T (known as maturity date).

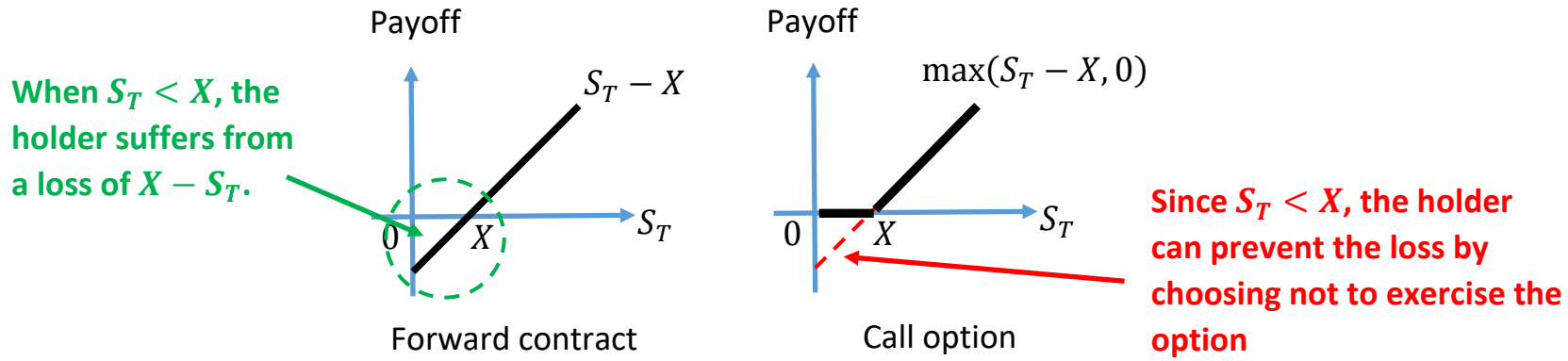
Classification of options

- There are two major types of options:
  - **Call option** – This gives option holder the right to *buy* the asset. The terminal payoff at time T is $\max(S_T - X, 0)$.
  - **Put option** – This gives option holder the right to *sell* the asset. The terminal payoff at time T is $\max(X - S_T, 0)$
- Each type of option is further divided into three categories:
  - **European option** -- the option holder can exercise the option at maturity date T only.
  - **American option** – the option holder can exercise the option at *any time prior to maturity date* T only.

Options versus forward contract

Different from the forward contract, the holder can decide whether to exercise the options depending on the profitability of the decision.

For example, we consider European call options and the corresponding forward contract (in long position) on an asset. The payoffs of these two derivatives are given as follows:



In other words, the options can provide holder a protection against the downside loss. In terms of value, the options should have larger value comparing with the corresponding forward contract.

Option price and option premium

Same as forward contract, the options allow the holder to buy (call)/sell (put) an asset at a fixed price X, regardless of the asset price at future price.

On the other hand, *the options provides holder protection against the downside loss that may occur in forward contract*. It is certainly unfavorable to the writer since he cannot gain anything now.

To compensate this, the writer requests additional upfront cost from the investors before the contract is formed. This explains why the option is more expensive than the forward counterpart. Such extra cost is called *insurance value*.

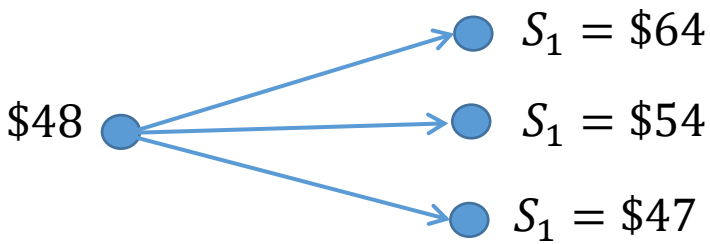
It is clear that the option price depends on the volatility of the asset price, which measures the occasion of the stock price, affects *positively* on the option price. If the volatility is high, the option can provide a larger protection to the investor and hence the option price is expected to be higher.

Can we determine the options price?

Unfortunately, there is no general pricing formula for bond options. Similar to the bond price and swap price, options price depends on the future cashflows received by holder. This depends on the future movement of the asset price.

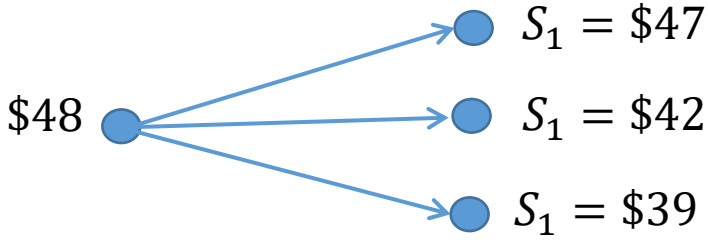
As an example, we consider a 1-year European call option on an asset with strike price \$50. We consider the following two "extreme" cases:

Case 1: The asset price is expected to be high in future



The option is always exercised at maturity date and a positive payoff is resulted, so the options price is high.

Case 2: The asset price is expected to be low in future



The option is never exercised at maturity date and the options price is simply 0.

Therefore, we need a *mathematical model* for asset price process in order to price the options.

Some general properties of options price

Although we cannot get a general pricing formula for options, it is still possible for us to study some general properties of options price. We summarize two basic properties below:

(*Note: For simplicity, we consider the case when *the asset does not pay any dividend* and the options is of European type.)

Property 1 (Put-call parity relation)

We let c_t and p_t be price of European call options and price of European put options at time t respectively. Suppose that both options have same strike price X and same maturity date T, then for any time $t \in [0, T]$, we have

$$c + XB(t; T) = p + S_t,$$

where S_t is the price of the asset at time t.

(*This basic equation describes the relation between call option and put option)

Property 2 (Options price with respect to strike price X)

We let $c(S_t; T, X)$ (resp. $p(S_t; T, X)$) be the prices of European call (resp. put) option with maturity date T and strike price X. Then we have

- *Monotonicity properties*
  Price of call option is decreasing with respect to X and the price of put option is increasing with respect to X: For any $X_1 < X_2$, we have
  $$c(S_t; T, X_1) > c(S_t; T, X_2)$$
  $$p(S_t; T, X_1) < p(S_t; T, X_2)$$
- *Convexity properties*
  Price of call option and price of put option are both strictly convex with respect to the strike price X. For any $X_1 < X_2$, $0 < \lambda < 1$, we let $X_0 = \lambda X_1 + (1 - \lambda)X_2$. We have
  $$c(S_t; T, X_0) < \lambda c(S_t; T, X_1) + (1 - \lambda)c(S_t; T, X_2)$$
  $$p(S_t; T, X_0) < \lambda p(S_t; T, X_1) + (1 - \lambda)p(S_t; T, X_2)$$

Proof of property 1

Both properties can be proved using no arbitrage pricing principle. We consider the following two portfolios:

Portfolio 1: Buy an European call option with strike price X and maturity date T and buy a zero coupon bond with face value X and maturity date T.

Portfolio 2: Buy an European put option with strike price X and maturity date T and buy one unit of the underlying asset.

Next, we compare the future payoffs of these two portfolios at time T. Since the options may or may not be exercised at time T (which depends on the asset price at that time), one needs to divide the analysis into two cases:

	$S_T \leq X$	$S_T > X$
Portfolio 1	$\underbrace{0}_{call} + \underbrace{X}_{bond} = \boldsymbol{X}$	$\underbrace{(S_T - X)}_{call} + \underbrace{X}_{bond} = \boldsymbol{S_T}$
Portfolio 2	$\underbrace{(X - S_T)}_{put} + \underbrace{S_T}_{asset} = \boldsymbol{X}$	$\underbrace{0}_{put} + \underbrace{S_T}_{asset} = \boldsymbol{S_T}$

We note that two portfolios generate the same future payoff at future time T. It follows from no arbitrage pricing principle that the current values of these two portfolios at time t must be the same. That is,

$$c + XB(t;T) = p + S_t.$$

Remark of the property 1

In fact, one can express the put-call parity using forward price. We let $F_{[t,T]}$ be the $T-t$ years forward price of the asset quoted at time t. Then we have

$$c + XB(t;T) = p + F_{[t,T]}B(t;T).$$

One can derive this equation by modifying the Portfolio 2 in the proof. That is, we replace the strategy "buy one unit of asset" by another strategy "entering into forward contract in long position with forward price $F_{[t,T]}$ and buy $T-t$ years zero coupon bond with face value $F_{[t,T]}$".

Since $F_{[t,T]} = \frac{S_t}{B(t;T)}$ if there is no arbitrage opportunity, we can deduce that

$$c + XB(t;T) = p + S_t.$$

Proof of property 2

We first prove the monotonicity property. We consider the following two portfolios:

Portfolio A: European call option with strike price X_1 and maturity date T

Portfolio B: European call option with strike price X_2 and maturity date T

Next, we compare the future payoff of these two portfolios at time T. We summarize the result in the following table:

	$S_T \leq X_1$	$X_1 < S_T \leq X_2$	$S_T > X_2$
Portfolio *A*	0	$S_T - X_1$	$S_T - X_1$
Portfolio *B*	0	0	$S_T - X_2$

One can observe that the value of Portfolio A is *always* strictly higher than that of Portfolio B. By no arbitrage pricing principle, the price of portfolio A at time t must be *higher* than that of Portfolio B. So we have

$$c(S_t, T; X_1) > c(S_t, T; X_2).$$

The monotonicity property for put option can be proved in a similar manner.

Next, we prove the convexity property of the option price. We consider the following two portfolios:

Portfolio C: An European call option with strike $X_0 = \lambda X_1 + (1-\lambda)X_2$.

Portfolio D: λ units of European call option with strike X_1 and $1-\lambda$ units of European call option with strike X_2.

Again, we proceed to compare their future values at time T:

	$S_T \le X_1$	$X_1 < S_T \le X_0$	$X_0 < S_T \le X_2$	$S_T > X_2$
Portfolio *C*	$\mathbf{0}$	$\mathbf{0}$	$\boldsymbol{S_T - X_0}$	$S_T - X_0$
Portfolio *D*	$\mathbf{0}$	$\boldsymbol{\lambda(S_T - X_1)}$	$\boldsymbol{\lambda(S_T - X_1)}$	$\boldsymbol{\lambda(S_T - X_1)}$ $\boldsymbol{+(1-\lambda)(S_T - X_2)}$ $\boldsymbol{= S_T - X_0}$

By simple algebra, we can establish that

$$S_T - X_0 = S_T - (\lambda X_1 + (1-\lambda)X_2)$$
$$\overset{S_T = \lambda S_T + (1-\lambda)S_T}{=} \lambda(S_T - X_1) + (1-\lambda)(S_T - X_2) \overset{S_T \le X_2}{\le} \lambda(S_T - X_1).$$

Thus the portfolio D has strictly higher payoff than portfolio C at maturity date T. By no arbitrage argument again, the portfolio value of C is smaller than that of D, i.e.

$$c(S_t, T; X_0) < \lambda c(S_t, T; X_1) + (1-\lambda)c(S_t, T; X_2).$$

Example 12 (A calculation example on put-call parity)

The market prices of various European options on a non-dividend paying asset are given in the following table:

	Type	Time-to-maturity	Strike price	Market price
Option A	Call	6 months	$50	$2.640
Option B	Call	1 year	$50	$2.915
Option C	Put	6 months	$50	$3.111

The spot rates of various maturities are given in the following table (*Note: All rates are quoted as annual effective yield rate)

	3 months	6 months	12 months	24 months
Spot rates	2%	2.2%	2.5%	3%

Assuming that there is no arbitrage opportunity, find the current price of 1-year European put option on the same asset with strike price $50.

😊 Solution

Without loss of generality, we take $t = 0$ to be current time.

We first determine the current price of the underlying asset. By applying the put-call parity on 1st and 3rd option, we have

$$\underbrace{2.640}_{c} + \underbrace{50\left(\frac{1}{(1.022)^{0.5}}\right)}_{XB(0;0.5)} = \underbrace{3.111}_{p} + S_0$$

$$\Rightarrow S_0 \approx 48.9879.$$

We let p_1 be the price of the 1-year put option. By applying put-call parity on 2nd option and this option, we get

$$\underbrace{2.915}_{c} + \underbrace{50\left(\frac{1}{(1.025)^{1}}\right)}_{XB(0;1)} = p_1 + \underbrace{48.9879}_{S_0}$$

$$\Rightarrow p_1 \approx 2.7076.$$

Example 13

The current price of a company stock is $S_t = \$75$. The following table shows the market information of two European options on this company stock:

Type	Time to maturity	Strike Price	Market Price
Call option	9 months	$73	$4.55
Put option	9 months	??	$2.82

You are given that the 9-month spot rate is 3%, quoted as annual nominal interest rate compounded continuously. We assume that the company stock does not pay any dividend in the coming 9 months

Question: What is the strike price of the put option?

(A) Higher than $73

(B) Equal to $73

(C) Lower than $73

😊Solution

> 😊IDEA: Without pricing formula, one cannot determine the exact value of X for this option. Instead, we can only deduce the *range* of X using the properties of option price.

We let $p(X)$ be the price of 9-month European put option with strike price X.

For our reference, we first determine the price of the 9-months European put option with strike price \$73. By applying put-call parity, we deduce that

$$p(73) + \underbrace{75}_{S_0} = \underbrace{4.55}_{c(73)} + \underbrace{73\left(\frac{1}{e^{0.03(0.75)}}\right)}_{XB(0;0.75)} \Rightarrow p(73) = 0.92584.$$

Since the market price of our target put option is higher than 0.92584 (i.e. $p(X) = 2.82 > 0.92584 = p(73)$) and the put option price should be increasing with respect to strike price, it follows that $X > 73$.

So the answer is **(A)**.

Identifying arbitrage opportunities using option price properties

Recall that all option properties are derived under the assumption that there is no arbitrage opportunity, hence one would expect that the arbitrage opportunity will exist if some of these properties are violated.

Example 14

The following table shows the market prices of 6-months European put options on a common asset.

	Time to maturity	Strike price	Market price
Option A	6 months	$40	$4.25
Option B	6 months	$45	$6.25
Option C	6 months	$50	$8.25

In other words, the option price forms a straight line (with positive slope) in (strike price)-(option price) plane. Identify an arbitrage opportunity.

😊Solution

To identify an arbitrage opportunity, we can check whether those option properties hold given the market data.

Under no arbitrage pricing principle, the put option price should be a convex function of strike price X. Note that $45 = \frac{1}{2}(40) + \frac{1}{2}(50)$, we expect that

$$\underbrace{p(45)}_{=6.25} < \underbrace{\frac{1}{2}p(40) + \frac{1}{2}(50)}_{=\frac{1}{2}(4.25)+\frac{1}{2}(8.25)=6.25}.$$

We observe that the inequality is violated and the put option with strike price \$45 appears to be overpriced. So one can construct an arbitrage opportunity using "buy low sell high" strategy as follows:

- Buy $\frac{1}{2}$ units of the put option with strike $\$40$ and $\frac{1}{2}$ units of the put option with strike $\$50$.
- Short-sell 1 unit of the put option with strike \$45.

Next, we shall argue that this trading strategy is an arbitrage opportunity by checking the conditions. The cost of setting this strategy is

$$Cost = \frac{1}{2}\underbrace{(4.25)}_{p(40)} + \frac{1}{2}\underbrace{(8.25)}_{p(50)} - \underbrace{(6.25)}_{p(45)} = 0.$$

So no cost is required in setting up this strategy.

Next, we compute the future payoff of this portfolio after 9 months:

	$S_T < 40$	$40 \le S_T < 45$	$45 \le S_T < 50$	$S_T \ge 50$
Future payoff	$\frac{1}{2}(40 - S_T)$ $+\frac{1}{2}(50 - S_T)$ $-(45 - S_T)$ $= \mathbf{0}$	$\frac{1}{2}(0)$ $+\frac{1}{2}(50 - S_T)$ $-(45 - S_T)$ $= \frac{1}{2}S_T - 20 \ge 0$	$\frac{1}{2}(0)$ $+\frac{1}{2}(50 - S_T)$ -0 $= \frac{1}{2}(50 - S_T) \ge 0$	$\frac{1}{2}(0) + \frac{1}{2}(0)$ $-(0)$ $= \mathbf{0}$

We observe the investor does not suffer from any potential loss and can earn some profit in some cases. So it is an arbitrage opportunity.

Example 15 (Harder)

You are given following information about marker prices of various European options on a non-dividend paying company stock.

Type	Time to maturity	Strike price	Market price
Call	6 months	$35	$1.28
Put	6 months	$38	$2

You are also given that

- The current term structure is flat and the annual nominal interest rate is 4% compounded continuously
- The current price of the stock is $38.6.

Identify an arbitrage opportunity.

☺IDEA

We first identify a relation between the relation between these options if there is no arbitrage opportunity.

😊Solution

We let $c(X)$ and $p(X)$ be the prices of 3-months European call options with strike X and 3-months European put options with strike X respectively.

Using put-call parity relation, the no-arbitrage price of call options with strike $\$38$ (if available in the market) is

$$\underbrace{c(38) + 38e^{-0.04\left(\frac{1}{2}\right)}}_{c+XB(t;T)} = \underbrace{p(38) + 38.6}_{p+S_t}$$

$$\Rightarrow c(38) \approx 3.35245.$$

The call options price should be *decreasing* with respect to strike price. So, we have

$$\underbrace{c(35)}_{=1.28} > \underbrace{c(38)}_{=3.35245} = p(38) + 38.6 - 38e^{-0.04\left(\frac{1}{2}\right)}.$$

However, the market data cannot fit the inequality properly. So there is an arbitrage opportunity. One can construct the trading strategy as follows:

- Buy (Long) 1 unit of call options with strike \$35.
- ~~Short 1 unit of call options with strike \$38.~~ (😵No!)

  Short 1 unit of put options with strike \$38, short 1 units of underlying assets (repay 1 unit of asset after 6 months) and buy a zero-coupon bond with face value 38.

The cost of constructing the portfolio is found to be

$$Cost = c(35) - p(38) - 38.6 + 38e^{-0.04\left(\frac{1}{2}\right)} \approx -2.07245.$$

So the investor receives an upfront payment \$2.07245 at time 0.

Next, we shall examine the terminal payoff of the portfolio after 6 months

	$S_T \leq 35$	$35 < S_T \leq 38$	$S_T > 38$
Total payoff	$0 - (38 - S_T)$ $-S_T + 38$ $\mathbf{= 0}$	$S_T - 35 - (38 - S_T)$ $- S_T + 38$ $\mathbf{= S_T - 35}$	$S_T - 35 - 0$ $-S_T + 38$ $\mathbf{= 3}$

We observe from the above table that the investor does not suffer any loss in future and may gain a positive profit when $S_T > 35$.

Hence, the trading strategy is an arbitrage opportunity.

Bond options

Bond options is an option on bond which allows the holder to buy/sell a bond (coupon bond or zero-coupon bond) at a specific price X on a specific date T (or a specific period). Although the bond options appear to be a type of ordinary option, it is useful in pricing many interest rate derivatives such as callable/puttable bond, interest rate cap, interest rate floor, swaption etc..

Basic properties of bond options (European type)

Similar to standard European options, the price of bond options observes the monotonicity properties and convexity properties (with respect to strike price X).

Put-call parity relation for bond options:

Since the underlying bond pays coupons at the intermediate time, the bond is a dividend paying asset. One needs to establish another version of put-call parity relation for bond options:

Properties 3 (Put-call parity for bond options)

We let c and p be the time-t price of T-year European call option and T-year European put option on a coupon bond respectively. The strike price of two options are both X. Suppose that the bond pays k coupons (of amount rF at times $T_1, T_2, \dots, T_k$ respectively, where $T_i \in (t, T)$, then

$$c + XB(t;T) = p + S_t - \underbrace{\sum_{i=1}^{k} rFB(t;T_i)}_{PV\ of\ coupons}.$$

Proof of property 3

We consider the following two portfolios:

Portfolio A: An European call bond option and buy T-year zero coupon bond with respect to X.

Portfolio B: An European put bond option and buy one unit of the underlying bond. In addition, for any $i = 1,2, \dots, k$, short a zero-coupon bond with face value rF and maturity date T_i.

(*Note: The coupon payment received from the bond is used to repay the face value rF from short-selling the bond. So the portfolio B will contain a put option and an underlying bond at time T.)

Then the future payoff of two portfolios at time T are given by

	$S_T \leq X$	$S_T > X$
Portfolio 1	$\underbrace{0}_{call} + \underbrace{X}_{bond} = \boldsymbol{X}$	$\underbrace{(S_T - X)}_{call} + \underbrace{X}_{bond} = \boldsymbol{S_T}$
Portfolio 2	$\underbrace{(X - S_T)}_{put} + \underbrace{S_T}_{asset} = \boldsymbol{X}$	$\underbrace{0}_{put} + \underbrace{S_T}_{asset} = \boldsymbol{S_T}$

Since the future payoffs of two portfolios are identical, then the current value of two portfolios must be the same. That is,

$$c + XB(t;T) = p + S_t - \sum_{i=1}^{k} rFB(t;T_i)$$

Example 16 (Put-call parity on bond option: A demonstration)

We consider a 5-year coupon bond issued by a company today. The bond has face value 100 and pays coupon semi-annually at an annual coupon rate 6%. The annual effective yield rate of the bond is currently 6%.

The following table shows the spot rates (quoted as annual effective interest rate) of various maturities:

	6 months	9 months	12 months	15 months
Spot rate	5.5%	5.6%	5.8%	6.10%

Suppose that the current price of 15-month European put option with strike price $100 on this bond is $5.86.

Calculate the price of 15-month European call option with strike price $100.

😊Solution

Firstly, the current price of the bond is found to be

$$S_0 = \sum_{i=1}^{10} \frac{100\left(\frac{0.06}{2}\right)}{(1.06)^{\frac{i}{2}}} + \frac{100}{(1.06)^5} = 100.3736.$$

On the other hand, two coupons have been paid during the first 15 months. Then the present value of these two coupons (denoted by D) is

$$D = \underbrace{3\left(\frac{1}{(1.055)^{\frac{1}{2}}}\right)}_{rFB(0;0.5)} + \underbrace{3\left(\frac{1}{(1.058)^1}\right)}_{rFB(0;1)} = 5.756293.$$

Using put-call parity, the price of the European call option (denoted by c) is

$$c + \underbrace{100\left(\frac{1}{(1.061)^{1.25}}\right)}_{XB(0;1.25)} = 5.86 + \underbrace{100.3736}_{S_0} - \underbrace{5.756293}_{D} \Rightarrow c \approx 7.611514.$$

Embedded bond options: Callable bond and puttable bond

In reality, bond options are frequently embedded in bonds when they are issued to make them more att ractive to either the issuer or potential investors. In this section, we shall explore two types of bonds with such features

Callable bond

Callable bond is a coupon bond which gives issuer a right to redeem the bond from holder at a specific price C at a specific period. Here, C is called *call price* (or *redemption price)*.

- The bond will be same as coupon bond if the bond is not called and the investor can receive the coupons and face value.
- Usually, the bond cannot be called during the first few years of the bond life (This period is known as *call protection period*).
- Call price is higher than the face value and is decreasing with respect to time. That is, the call price is normally lower if the bond is called at later date.

Use of callable bond

Firstly, callable bond allows the company (issuer) to adjust its capital structure. In particular, it can redeem the bond earlier (early repayment) to improve its debt-equity ratio.

Secondly, callable bond allows the company to refinance the debt at a lower interest rate when the market interest rate drops in future.

Example 17 (Application of callable bond in debt management)

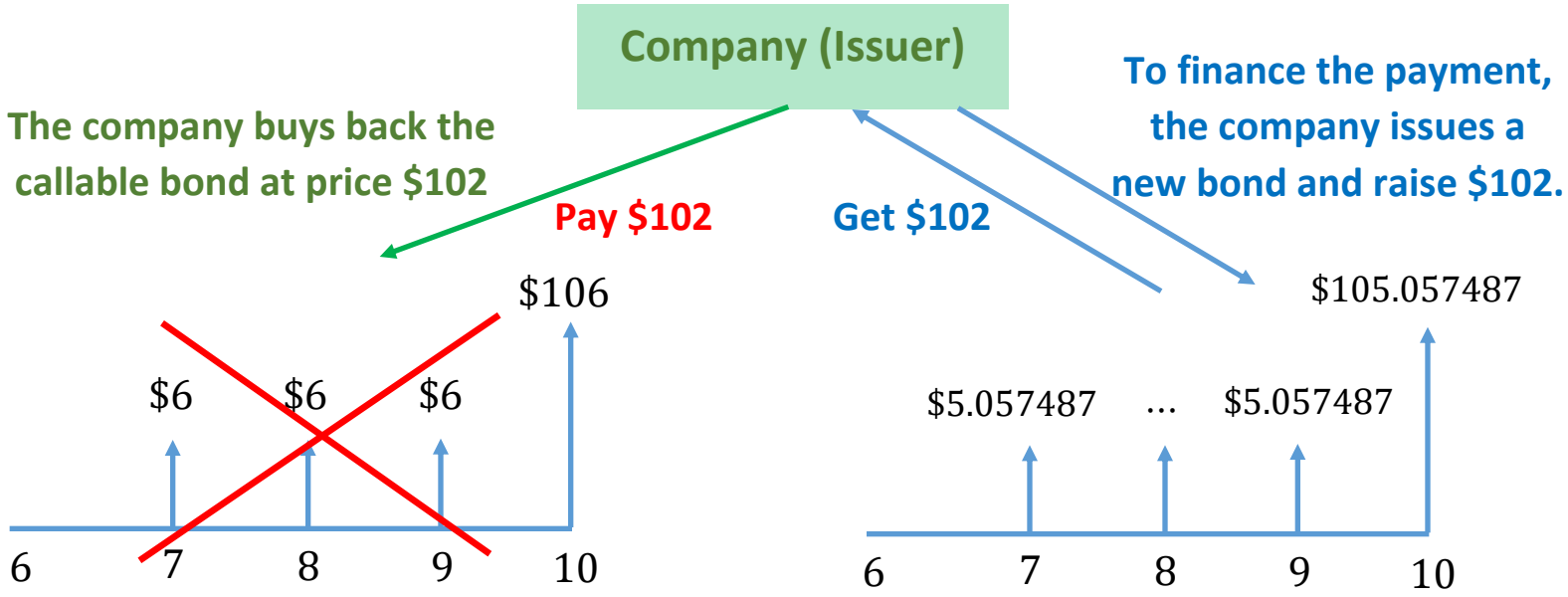
As an example, we consider a 10-year callable bond issued by a company. The bond has face value 100 and annual coupon rate 6% payable annually.

- The company can redeem the bond starting from 6th coupon payment.
- The call price is **$102** if the bond is called before maturity and is **$100** if it is called at maturity.
- For simplicity, we assume that the term structure of interest rate is always flat. That is, $s_0(T) = c$ for all $T > 0$.

Suppose that the spot rate drops to 4.5% at the end of 6th year (just after 6th coupon payment), the present value of the remaining coupons and face value is

$$PV = \sum_{i=1}^{4} \frac{100(0.06)}{(1.045)^i} + \frac{100}{(1.045)^4} \approx 105.3813 > 102.$$

Then the company can reduce its repayment by calling the original bond (at \$102) and issuing a new bond:



Pricing of callable bond

Comparing with standard coupon bond, the issuer also has a right to redeem the bond at a certain period of time. In other words, the issuer is holding a call option (Bermudan type) on this coupon bond.

When the bondholder buys this coupon bond, he/she is also selling the call bond option to the issuer. Hence, the price of this callable bond can be expressed as

$$Price\ of\ callable\ bond = Price\ of\ coupon\ bond - Price\ of\ bond\ option\ (call).$$

Remark

- Since the price of bond option is positive in general, it implies that the price of callable bond is lower than that of non-callable counterpart (i.e. standard coupon bond).
- Hence, the yield rate of the callable bond should be *higher* than that of corresponding standard coupon bond.

Puttable bond

The puttable bond gives the *bondholder* the right, but not obligation, to demand an early redemption at a specific price at a certain period of times in future. That is, request the issuer to pay the redemption value at some time before the maturity date of the bond.

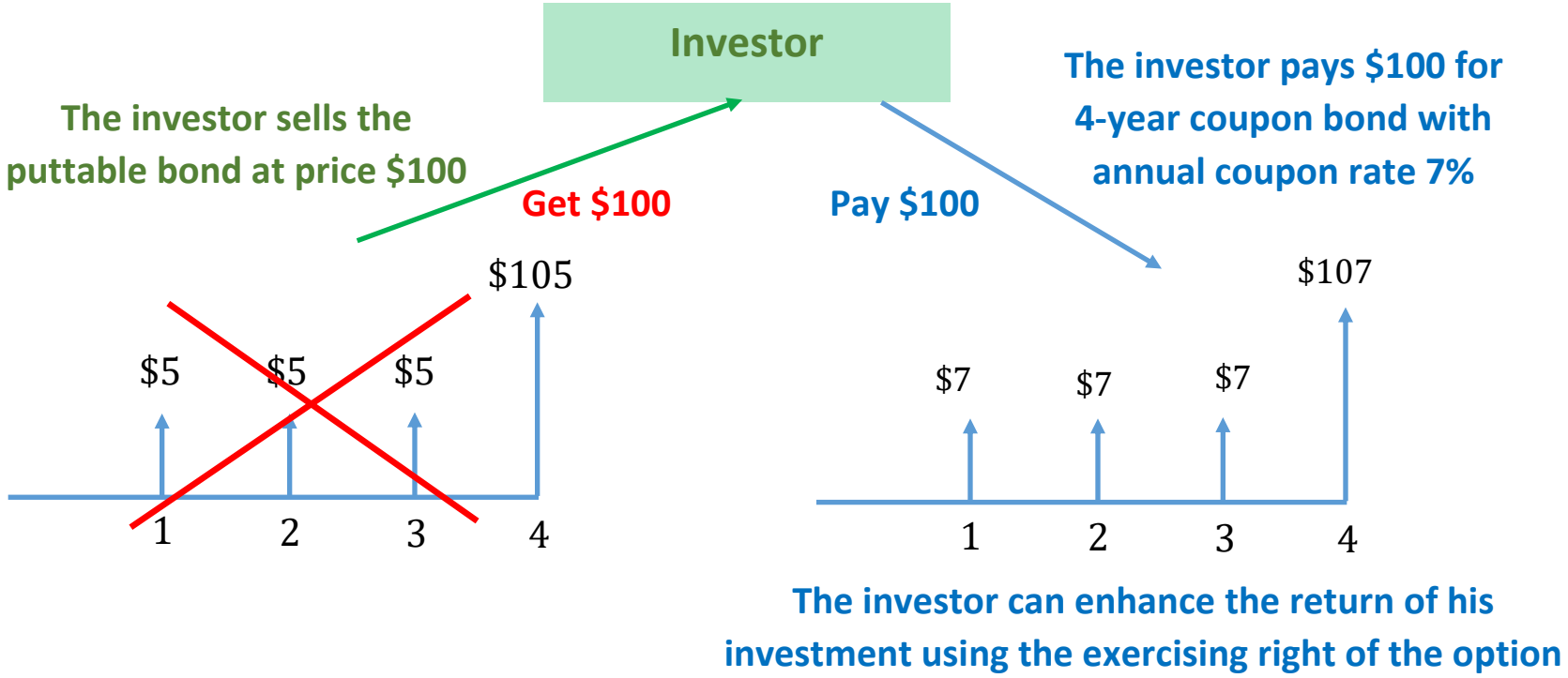
At investor's point of view, the puttable bond allows the bondholder to get back his/her investment capital earlier and reinvest when the interest rate goes high in future.

Example 18

The investor is holding a 4-year puttable bond. The bond has face value \$100 and annual coupon rate 5% payable annually. *The bond allows the investor to sell the bond to the issuer at price \$100 at any coupon payment date prior to the maturity*. The term structure is assumed to be flat and the annual effective interest rate is currently $s_0(T) = 5.2\%$.

Suppose that the interest rate climbs to 7%, the present value of the coupons and face value becomes $\sum_{i=1}^{4} \frac{5}{(1.07)^i} + \frac{100}{(1.07)^4} \approx 93.22558 < 100.$

Instead of keeping the bond, the investor can exercise his "option" immediately and sell the bond for \$100. Then investor can invest the amount received into the coupon bond with higher coupon rate (e.g. 4-year fixed rate coupon bond with annual coupon rate 7% or 4-year floating rate note etc.)



The investor can enhance the return of his investment using the exercising right of the option

Pricing of puttable bond

Comparing with the non-puttable counterpart (i.e. standard coupon bond), the bondholder has a right to sell the bond at a fixed price K. Besides the coupon bond, the bondholder also acquires a put option (Bermudan type) on this coupon bond when he buys the puttable bond.

Using no arbitrage pricing principle, the current price of the puttable bond can be expressed as

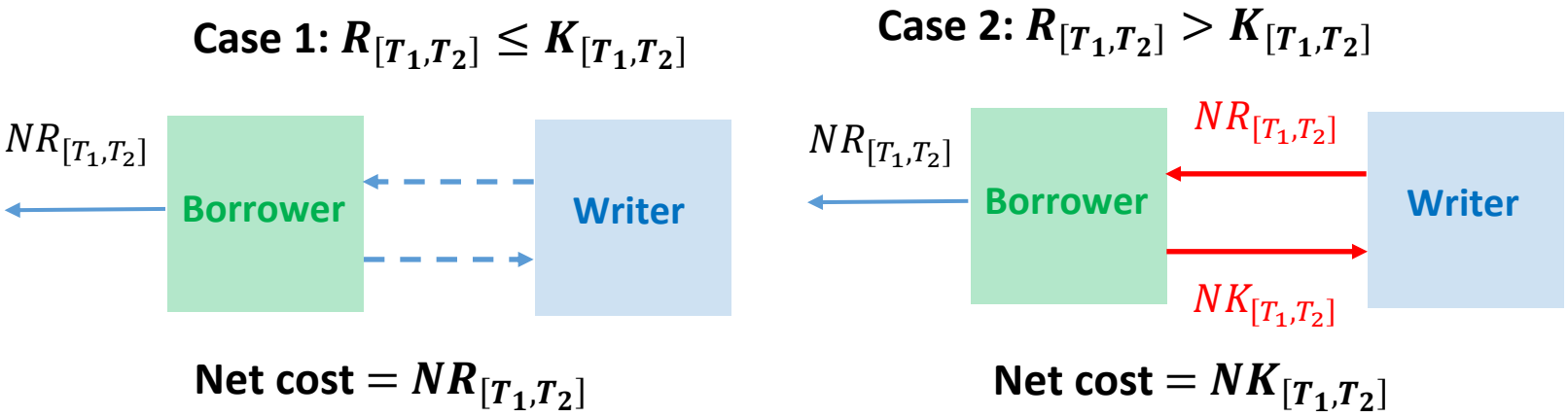
$$Price\ of\ puttable\ bond = Price\ of\ coupon\ bond + Price\ of\ bond\ option\ (put).$$

- Since the price of the put option is non-negative (or positive in most of the cases), so the price of the puttable bond is always higher than that of the standard coupon bond (non-puttable).
- As a result, the yield rate of the puttable bond will be *lower* than that of the corresponding standard coupon bond.

Options on interest rate -- Interest rate caplet and interest rate floorlet

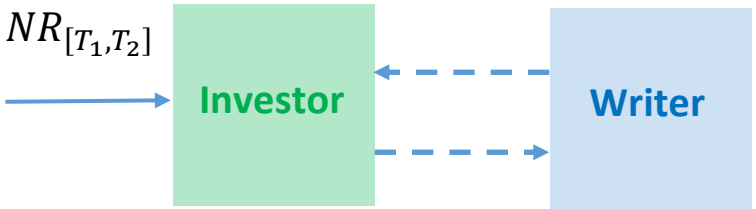
Interest rate caplet/floorlet is an interest rate derivative which allows investor to hedge against the potential interest rate risk in future.

- **Interest rate caplet** – It is used by the borrower to "cap" the borrowing cost of his short-term floating rate loan over the future period $[T_1, T_2]$. By holding the interest rate caplet, the borrower will receive a payment of $NR_{[T_1,T_2]} - NK_{[T_1,T_2]}$ at time T_2 if the interest rate $R_{[T_1,T_2]}$ is above the fixed rate $K_{[T_1,T_2]}$ (also known as strike rate) and receive nothing if otherwise. So the borrowing rate of the loan will be capped at $K_{[T_1,T_2]}$.



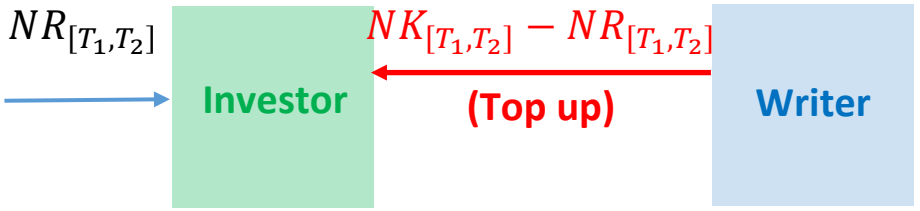
- **Interest rate floorlet** – It is used by investor who will engage in an investment that yield floating rate of return $R_{[T_1,T_2]}$ in future to "control" the return of his investment.
  By holding the interest rate floorlet, the investor will receive a payment of $NK_{[T_1,T_2]} - NR_{[T_1,T_2]}$ at time T_2 if the interest rate $R_{[T_1,T_2]}$ is below the fixed rate $K_{[T_1,T_2]}$ (also known as strike rate) and receive nothing if otherwise. Consequently, the minimum return rate earned by the investment is $K_{[T_1,T_2]}$.

Case 1: $\boldsymbol{R_{[T_1,T_2]} \geq K_{[T_1,T_2]}}$



Net gain $= \boldsymbol{NR_{[T_1,T_2]}}$

Case 2: $\boldsymbol{R_{[T_1,T_2]} < K_{[T_1,T_2]}}$



Net gain $= \boldsymbol{NK_{[T_1,T_2]}}$

(*Note: Here, the notional principal N represents the amount of capita invested into the investment)

Interest rate caplet/floorlet v.s. European options

- **Interest rate caplet**

  Note that the terminal payoff of the interest rate caplet at time T_2 is

$$Terminal\ payoff = \begin{cases} NR_{[T_1,T_2]} - NK_{[T_1,T_2]} & if\ \ R_{[T_1,T_2]} > K_{[T_1,T_2]} \\ 0 & if\ \ R_{[T_1,T_2]} \leq K_{[T_1,T_2]} \end{cases}$$

$$= \max\left(NR_{[T_1,T_2]} - NK_{[T_1,T_2]}, 0\right).$$

  In other words, the caplet can be seen as an European *call* option on floating rate payment with "strike price" $NK_{[T_1,T_2]}$, where N is the notional principal.

- **Interest rate floorlet**

  Similarly, one can show that the terminal payoff of the floorlet is

$$Terminal\ payoff = \max\left(NK_{[T_1,T_2]} - NR_{[T_1,T_2]}, 0\right).$$

  So floorlet can be seen as European *put* option on floating rate payment with strike price $NK_{[T_1,T_2]}$.

Remark: Quotation of interest rate caplet and interest rate floorlet

We called such interest rate caplet/floorlet as "T_2-year interest rate caplet/floorlet on $(T_2 - T_1)$ year spot rate/LIBOR rate with notional principal N and strike rate $K_{[T_1,T_2]}$", where

- N denotes the notional principal;
- T_2 denotes the maturity date of the options;
- "$(T_2 - T_1)$ year spot rate" denotes the (effective) interest rate applied over the period $[T_1, T_2]$, i.e. $R_{[T_1,T_2]}$.
- $K_{[T_1,T_2]}$ is the fixed (effective) interest rate applied over the period $[T_1, T_2]$.

(*Note: In practice, both spot rate and strike rate are quoted as annual rate. In many financial textbooks, both rates are often quoted as *annual nominal interest rate with compounded frequency that matches the length of the period* $[T_1, T_2]$. In this case, $R_{[T_1,T_2]}$ and $K_{[T_1,T_2]}$ are computed as

$$R_{[T_1,T_2]} = R^{Annual}_{[T_1,T_2]}(T_2 - T_1), \qquad K_{[T_1,T_2]} = K^{Annual}_{[T_1,T_2]}(T_2 - T_1).)$$

Pricing of interest rate caplet and floorlet

Since both caplet and floorlet are options-like derivatives, the pricing of these options requires a model which simulate the interest rate movement in future. Nevertheless, we can express the pricing formula of these options in terms of prices of bond option using no arbitrage pricing principle.

Pricing formula of interest rate caplet (A little bit technical)

In order to price the caplet, one has to find a portfolio which the future payoff is identical to that of the caplet. This can be done by a two-step procedure.

- ✓ Recall that the terminal payoff of the interest rate caplet at time T_2 (denoted by V_{T_2} is $\max\left(NR_{[T_1,T_2]} - NK_{[T_1,T_2]}, 0\right)$.
  Since the floating rate $R_{[T_1,T_2]}$ is known at time T_1, so the value of V_{T_2} is also known at time T_1. Therefore, the present value of V_{T_2} at time T_1 (denoted by V_{T_1}) is

$$V_{T_1} = \frac{\max\left(NR_{[T_1,T_2]} - NK_{[T_1,T_2]}, 0\right)}{1 + R_{[T_1,T_2]}} = \max\left(\frac{NR_{[T_1,T_2]} - NK_{[T_1,T_2]}}{1 + R_{[T_1,T_2]}}, 0\right).$$

✓ By some algebra, the payoff V_{T_1} can be expressed as

$$V_{T_1} = \max\left(\frac{NR_{[T_1,T_2]} - NK_{[T_1,T_2]}}{1 + R_{[T_1,T_2]}}, 0\right)$$

$$= \max\left(\frac{N\left(1 + R_{[T_1,T_2]}\right) - N\left(1 + K_{[T_1,T_2]}\right)}{1 + R_{[T_1,T_2]}}, 0\right)$$

$$= \max\left(\underbrace{N}_{\substack{strike \\ price}} - \underbrace{\frac{N\left(1 + K_{[T_1,T_2]}\right)}{1 + R_{[T_1,T_2]}}}_{"bond\ price"}, 0\right)$$

This term can be treated as the time-T_1 price of zero coupon bond with face value $N\left(1 + K_{[T_1,T_2]}\right)$ and maturity date T_2.

Thus the payoff V_{T_1} is same as the terminal payoff of a T_1-year bond put option on a T_2-year zero coupon bond with face value $N\left(1 + K_{[T_1,T_2]}\right)$ with strike price N.

✓ To price the caplet, we consider the following portfolio:

- Buy a bond put option mentioned above. The investor will receive a payoff of $\max\left(N - \frac{N\left(1+K_{[T_1,T_2]}\right)}{1+R_{[T_1,T_2]}}, 0\right)$ at time T_1.
- At time T_1, we reinvest the amount received at interest rate $R_{[T_1,T_2]}$ so that we can receive an amount of $\left(1 + R_{[T_1,T_2]}\right)\max\left(N - \frac{N\left(1+K_{[T_1,T_2]}\right)}{1+R_{[T_1,T_2]}}, 0\right) = \max\left(NR_{[T_1,T_2]} - NK_{[T_1,T_2]}, 0\right)$ at time T_2.

Since the future payoff of the portfolio matches that of the caplet, it follows from no arbitrage pricing principle that

$$\left\{\begin{matrix}Current\ price\ of \\ interest\ rate\ caplet\end{matrix}\right\} = \left\{\begin{matrix}Current\ price\ of\ T_1 - year\ put\ option\ on\ T_2 - year \\ zero\ coupon\ bond\ with\ face\ value\ N\left(1 + K_{[T_1,T_2]}\right)\end{matrix}\right\}.$$

Pricing formula of interest rate floorlet

Similarly, one can use the similar method (left as exercise) and deduce that

$$\left\{\begin{matrix}Current\ price\ of \\ interest\ rate\ floorlet\end{matrix}\right\} = \left\{\begin{matrix}Current\ price\ of\ T_1 - year\ call\ option\ on\ T_2 - year \\ zero\ coupon\ bond\ with\ face\ value\ N\left(1 + K_{[T_1,T_2]}\right)\end{matrix}\right\}.$$

Example 19

We consider a 3-year interest rate floorlet on 6-month spot rate. The notional principal is \$1000. The strike rate of the floorlet is $K = 6\%$, which is quoted as annual nominal interest rate compounded semi-annually. Express the price of interest rate floorlet in terms of price of some bond options.

☺Solution

Note that the payoff of the floorlet at **time 2.5** can be expressed as

$$V_{2.5} = \frac{1}{1 + R_{[2.5,3]}} \max\left(NK_{[2.5,3]} - NR_{[2.5,3]}, 0\right)$$

$$= \max\left(\frac{NK_{[2.5,3]} - NR_{[2.5,3]}}{1 + R_{[2.5,3]}}, 0\right) = \max\left(\frac{N\left(1 + K_{[2.5,3]}\right)}{1 + R_{[2.5,3]}} - N, 0\right)$$

$$\overset{N=1000, K_{[2.5,3]}=\frac{0.06}{2}}{=} \max\left(\frac{1030}{1 + R_{[2.5,3]}} - 1000, 0\right),$$

which is a terminal payoff of 2.5-year bond call option on 3-year zero coupon bond with face value 1030 and the strike price is 1000. By no arbitrage pricing principle, the price of the floorlet equals to the price of this bond call option.

Example 20 (A bit harder)

The following table shows the market price of various European options on a 2-year zero coupon bond with face value 100.

	Type	Time to maturity	Strike price	Market price
Option A	Call	1 years	$95	$2.8
Option B	Call	1 years	$98	$1.3
Option C	Call	1.5 years	$95	$5.2
Option D	Call	1.5 years	$98	$2.9
Option E	Put	1 years	$95	$3.43
Option F	Put	1 years	$98	$4.788
Option G	Put	1.5 years	$95	$3.4
Option H	Put	1.5 years	$98	$3.88

Assuming that there is no arbitrage opportunity in the market, find the market prices of each of the following options:

(a) 2-year interest rate caplet on 1-year spot rate with strike rate 5.263% and notional principal $N = \$5000$.

(b) 2-year interest rate floorlet on 6-month spot rate with strike rate 4.1216% and notional principal $N = \$10000$.

Here, two strike rates are quoted as annual effective interest rate.

😊Solution

(a) We should find the price using no arbitrage pricing principal. Since the terminal payoff of the caplet at time 2 is

$$V_2 = \max\left(NR_{[1,2]} - NK_{[1,2]}, 0\right),$$

where $K_{[1,2]} = 0.05263$.

Thus, the value of the caplet at time 1 can be expressed as

$$V_1 = \frac{1}{1 + R_{[1,2]}} \max\left(NR_{[1,2]} - NK_{[1,2]}, 0\right) = \max\left(\frac{NR_{[1,2]} - NK_{[1,2]}}{1 + R_{[1,2]}}, 0\right)$$

$$= \max\left(N - \frac{N\left(1+K_{[1,2]}\right)}{1+R_{[1,2]}}, 0\right)$$

$$= \frac{N\left(1+K_{[1,2]}\right)}{100}\max\left(\frac{100}{1+K_{[1,2]}} - \frac{100}{1+R_{[1,2]}}, 0\right)$$

$$\overset{\substack{N=5000 \\ K_{[1,2]}=5.263\%}}{=} 52.6315 \underbrace{\max\left(95 - \frac{100}{1+R_{[1,2]}}, 0\right)}_{put\ option\ with\ strike\ \$95}.$$

So the caplet can be replicated by buying 52.6315 units of the 1-year put option with strike price \$95.

Using no arbitrage pricing principle, the current price of the interest rate caplet is

$$V_0 = 52.6315 \times \$3.43 \approx 180.526.$$

(b) Using similar approach, we consider the value of the floorlet at time 1.5:

$$V_1 = \frac{1}{1+R_{[1.5,2]}}\max\left(NK_{[1.5,2]} - NR_{[1.5,2]}, 0\right) = \max\left(\frac{NK_{[1.5,2]} - NR_{[1.5,2]}}{1+R_{[1.5,2]}}, 0\right)$$

$$= \max\left(\frac{N\left(1+K_{[1.5,2]}\right)}{1+R_{[1.5,2]}} - N, 0\right)$$

$$= \frac{N\left(1+K_{[1.5,2]}\right)}{100}\max\left(\frac{100}{1+R_{[1.5,2]}} - \frac{100}{1+K_{[1.5,2]}}, 0\right)$$

$$\overset{\substack{N=10000 \\ K_{[1.5,2]}=(1.041216)^{\frac{1}{2}}-1=0.0204}}{=} 102.04 \underbrace{\max\left(\frac{100}{1+R_{[1.5,2]}} - 98, 0\right)}_{call\ option\ with\ strike\ \$98}.$$

Therefore, the floorlet can be replicated by buying 102.04 units of 1.5-years bond call option with strike \$98.

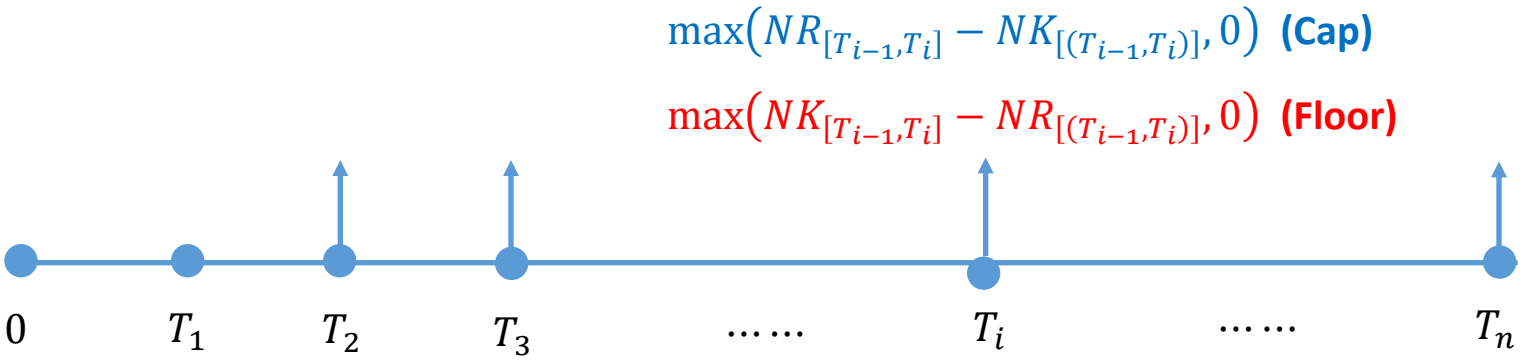
By no arbitrage pricing principle, the current price of the floorlet is

$$V_0 = 102.04 \times 2.9 = 295.916.$$

Interest rate caps and interest rate floors

Interest rate cap is a derivative which the holder will receive an amount of $\max\left(NR_{[T_{i-1},T_i]} - NK_{[T_{i-1},T_i]}, 0\right)$ at time $T_i = iT$, where $i = 2,3,\ldots,n$.

On the other hand, holder of the interest rate floor will receive $\max\left(NK_{[T_{i-1},T_i]} - NR_{[T_{i-1},T_i]}, 0\right)$ at time $T_i = iT$, where $i = 2,3,\ldots,n$

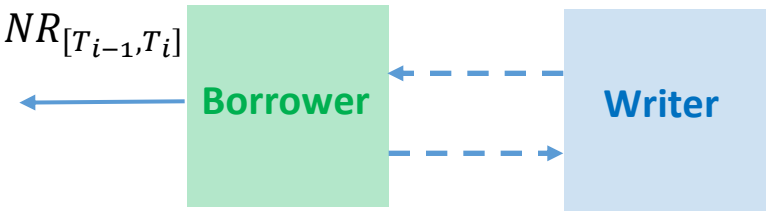


- Note that there is no payment at time T_1, it is because the interest rate $R_{[0,T_1]}$ is fixed at time 0 (current time) and there is no optionality.
- Here, the fixed rate $K_{[T_{i-1},T_i]} = K$. K is called *cap rate* (for interest rate cap) or *floor rate* (for interest rate floor).

Use of interest rate cap and interest rate floor

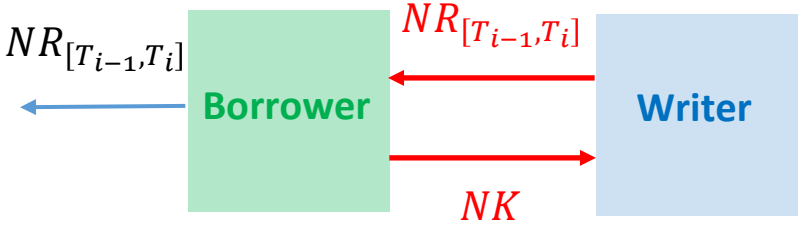
Similar to interest rate caplet, interest rate cap is often used by borrower who needs to make *multiple floating interest payment periodically*. By holding the interest rate cap, the borrower is protected against the borrowing rate raises above a certain level K (i.e. borrowing rate is at most K)

Case 1: $R_{[T_{i-1},T_i]} \leq K$



When the interest rate is low which $R_{[T_{i-1},T_i]} \leq K$, the options is not exercised (and no payout from the cap) and the borrower pays $NR_{[T_{i-1},T_i]}$.

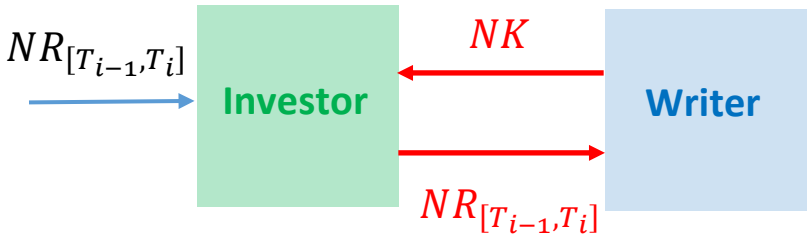
Case 2: $R_{[T_{i-1},T_i]} > K$



When $R_{[T_{i-1},T_i]} > K$, the options is exercised and the borrower gets a payoff of $NR_{[T_{i-1},T_i]} - NK$ from the options. As a result, the borrower just need to pays NK.

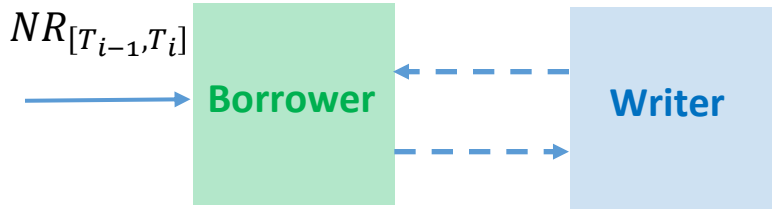
Interest rate floor can be used by investors who engage into some investment which earns floating rate of return periodically. By purchasing interest rate floor, the investor can guarantee a minimum return K from the investment.

Case 1: $R_{[T_{i-1},T_i]} < K$



When the interest rate is low which $R_{[T_{i-1},T_i]} < K$, the options is exercised and the investor gets a payoff of $NK - NR_{[T_{i-1},T_i]}$ to top up the return. As a result, the investor earns NK.

Case 2: $R_{[T_{i-1},T_i]} \geq K$



When the interest rate is high which $R_{[T_{i-1},T_i]} \geq K$, the options is not exercised (and no payout from the cap) and the investor earn $NR_{[T_{i-1},T_i]}$.

Pricing formula for interest rate cap and floor

- One can observe that the interest rate cap is simply a portfolio consists of a series of caplets on spot rate with various maturities $(T_2, T_3, \dots, T_n)$. Using no arbitrage pricing principle, the price of the interest rate cap is simply the sum of the prices of these caplets. Similarly, the price of the interest rate floor is the sum of the prices of the corresponding floorlets.
- (Put-call parity relation for interest rate swap and caplet)
  In fact, one can establish the following relationship between interest rate cap and interest rate floor:

$$\begin{Bmatrix} Price\ of\ interest \\ rate\ cap \end{Bmatrix} = \begin{Bmatrix} Price\ of\ interest \\ rate\ floor \end{Bmatrix} + \begin{Bmatrix} Price\ of\ interest \\ rate\ swap\ (floating \\ rate\ receiver) \end{Bmatrix}.$$

  Here, the interest rate swap allows holder to receive the floating rate payment $NR_{[T_{i-1},T_i]}$ and pay the fixed rate payment NK at times $T_2, T_3, \dots, T_n$ and no exchange is made at time T_1.

To see this, we consider the following two portfolios:

Portfolio 1: Buy an interest rate cap with strike rate K

Portfolio 2: Buy an interest rate floor with strike rate K and interest rate swap (as floating rate receiver) with swap rate K.

For any $i = 2,3, \dots, n$, the payoffs of these two portfolios at time T_i are given as follows:

	$R_{[T_{i-1},T_i]} < K$	$R_{[T_{i-1},T_i]} \geq K$
Portfolio 1	$\mathbf{0}$	$\boldsymbol{R_{[T_{i-1},T_i]} - K}$
Portfolio 2	$K - R_{[T_{i-1},T_i]} + \left(R_{[T_{i-1},T_i]} - K\right) = \mathbf{0}$	$0 + \left(R_{[T_{i-1},T_i]} - K\right) = \boldsymbol{R_{[T_{i-1},T_i]} - K}$

Since the future payoffs of two portfolios are the same, it follows from no arbitrage pricing principle that the current values of these two portfolio are the same. That is,

$$\underbrace{\left\{\begin{matrix}Price\ of\ interest\\ rate\ cap\end{matrix}\right\}}_{Portfolio\ 1} = \underbrace{\left\{\begin{matrix}Price\ of\ interest\\ rate\ floor\end{matrix}\right\} + \left\{\begin{matrix}Price\ of\ interest\\ rate\ swap\ (floating\\ rate\ receiver)\end{matrix}\right\}}_{Portfolio\ 2}.$$

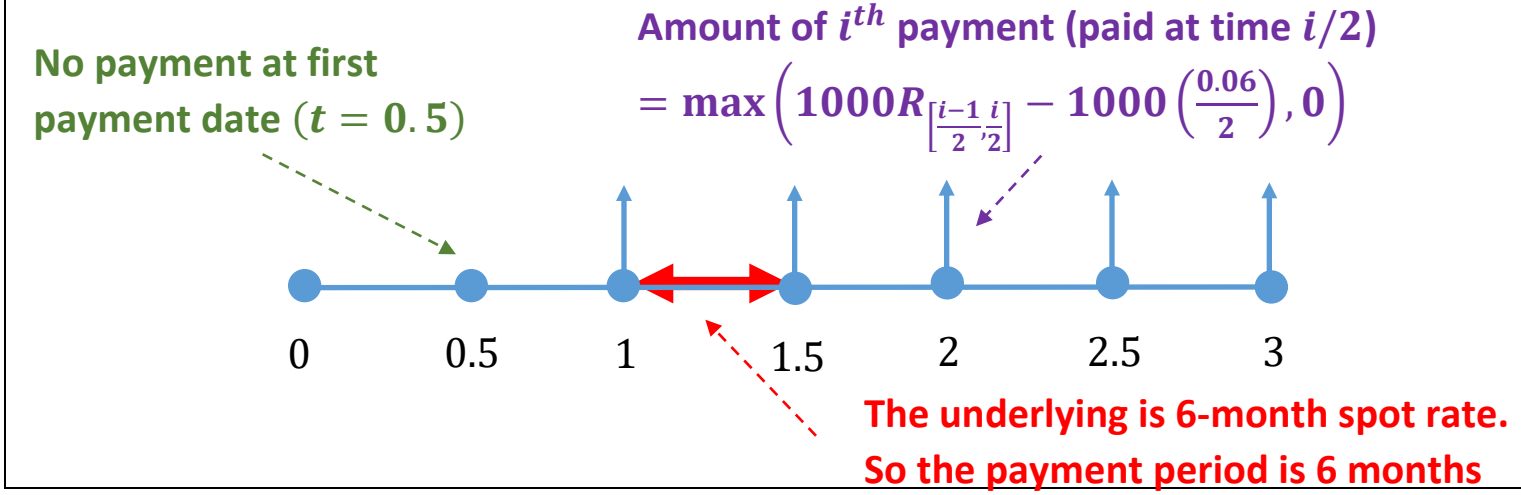
Example 21 (Pricing of interest rate cap using bond options)

We consider a 3-year interest rate cap on 6-month spot rates issued today. The notional principal is 1000 and the strike rate is 6% which is quoted as annual nominal interest rate compounded semi-annually. We let

- $p(0;T,T_B,X)$ be the current price of T-year bond put option with strike price X on T_B-year zero coupon bond with face value 1000.

Express the current price of the interest rate cap in terms of $p(T,T_B,X)$.

Remark: The cashflow of the 3-year interest rate cap is described as follows:



😊Solution

We first compute the current price of caplet which paid $\max\left(1000R_{\left[\frac{i-1}{2},\frac{i}{2}\right]} - 1000\left(\frac{0.06}{2}\right), 0\right)$ at time $\frac{i}{2}$, where $i = 2,3,\ldots,6$.

Using similar method as in earlier example, we consider the caplet value at time $\frac{i-1}{2}$:

$$V_{\frac{i-1}{2}} = \frac{1}{1+R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}}\max\left(1000R_{\left[\frac{i-1}{2},\frac{i}{2}\right]} - 1000\left(\frac{0.06}{2}\right), 0\right)$$

$$= \max\left(1000 - \frac{1030}{1+R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}}, 0\right) = \frac{1030}{1000}\max\left(\frac{1000000}{1030} - \frac{1000}{1+R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}}, 0\right)$$

$$= 1.03\underbrace{\max\left(970.8738 - \frac{1000}{1+R_{\left[\frac{i-1}{2},\frac{i}{2}\right]}}\right)}_{\frac{i-1}{2}\ year\ put\ option\ on\ \frac{i}{2}-year\ zero\ coupon\ bond\ with\ face\ value\ 100.}.$$

So the caplet can be replicated by holding 1.03 units of $\frac{i-1}{2}$ years put option (with strike price 970.8738) on $\frac{i}{2}$ year zero coupon bond with face value 100.

By no arbitrage pricing principle, the current price of the caplet should match the price of the corresponding bond put option, i.e. $p\left(0;\frac{i-1}{2},\frac{i}{2},970.8738\right)$.

Since the interest rate cap can be seen as a portfolios of 5 independent caplets, it follows from no arbitrage pricing principle that the current price of the interest rate cap is the sum of the prices of these 5 caplets. That is,

$$V_0 = \sum_{i=2}^{6} 1.03 \left\{ \begin{matrix} Price\ of\ \frac{i}{2}\ years\ interest\ rate\ caplets\ on \\ 6-month\ spot\ rates\ with\ strike\ 970.8738 \end{matrix} \right\}$$

$$= 1.03 \sum_{i=2}^{6} p\left(0;\frac{i-1}{2},\frac{i}{2},970.8738\right).$$

(*If the prices/ pricing formula of the bond put options are given, the price of the cap can be computed using the above formula.)

Example 22 (Price of on-going interest rate cap)

We consider **Example 21** again. Suppose that 15 months has passed, the 6-month spot rates observed at time 1 is 6% and the 3-month spot rate is currently 5.5% (both rates quoted as annual effective interest rate). We let $p(1.25; T, T_B, X)$ be the current price of T-year bond call option with strike price X on zero coupon bond with face value 1000 and maturity date T_B. Find the new price of the interest rate cap.

😊Solution

Note that the amount of next payment (paid at time 1.5) is

$$\max\left(\underbrace{1000\left((1.06)^{\frac{1}{2}}-1\right)-1000(0.03)}_{1000R_{[1,1.5]}-1000K},0\right)=\max(-0.43,0)=0.$$

So the new price of the cap is given by

$$V_0=\underbrace{(0)B(1.25;1.5)}_{PV\ of\ next\ payment}+1.03\sum_{i=4}^{6}p\left(1.25;\frac{i-1}{2},\frac{i}{2},970.8738\right).$$

Swaption – Options on swaps

Swaption is an option which gives holder the right to enter a certain position of an interest rate swap with a specific swap rate K and notional principal N at a specified future time T. Typically, there are two types of swaptions:

- **Receiver swaption** (Call options) – the holder has the right to enter into the swap as *fixed rate receiver*.
- **Payer swaption** (Put options) – the holder has the right to enter into the interest rate swap as *fixed rate payer*.

An application of swaption

Swaption (payer swaption) can be used by the company to limit the borrowing cost (interest rate payment) at some level K if it wishes to borrow money at future time and faces an uncertainty in future movement of interest rate.

- ✓ The company can enjoy a low interest payment when the interest rate stays low in future.
- ✓ The company acquire protection from the swaption when the future interest rate raises.

Terminal payoff of the swaption (European style)

The swaption holder needs *not* to pay any cost when exercising the options. Let V_T^{Swap} be the value of the underlying interest rate swap at maturity time T.

Since the swap rate K of the underlying swap is fixed at time 0 and the future interest rate may change, it is very likely that the value of the swap is non-zero although it is initialized at time T.

Intuitively, one would exercise the option and enter into the swap only when the swap value is positive, i.e. V_T^{Swap}. Thus the terminal payoff of the swaption can be expressed as

$$Terminal\ payoff = \begin{cases} V_T^{Swap} & if\ \ V_T^{Swap} > 0 \\ 0 & if\ otherwise \end{cases} = \max\left(V_T^{Swap}, 0\right).$$

Mathematically, one can argue as follows: As an example, we consider the receiver swaption (i.e. the holder will act as fixed rate receiver of the swap when he exercises the swaption)

Given the swap rate of the interest rate swap quoted at time T (denoted by K_T), then the value of the underlying interest rate swap (at fixed rate *receiver* side) can be expressed as

$$V_T = \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ fixed\ rate\ K \end{Bmatrix} - \begin{Bmatrix} PV\ of\ floating \\ leg\ payments \end{Bmatrix}$$

$$= \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ fixed\ rate\ K \end{Bmatrix} - \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ fixed\ rate\ K_T \end{Bmatrix}$$

$$= \begin{Bmatrix} PV\ of\ fixed\ leg\ payments \\ with\ fixed\ rate\ (K - K_T) \end{Bmatrix} = \sum_{i=1}^{n} N(K - K_T)B(T; T + iT_C)$$

(Here, $T + iT_c$ is the exchange dates of the interest rate swap)

If $V_T \geq 0$, we have $K > K_T$. The holder finds himself better off to enter into the underlying interest rate swap to enjoy higher fixed rate payment and he will exercise the options

If $V_T < 0$, we have $K < K_T$. The holder will not exercise the options since it is better for him to enter into interest rate swap in the market with swap rate K_T.

Pricing of the swaption using bond options

Similar to the case for interest rate cap and interest rate floor, we would like to obtain the pricing formula of the swaption using bond options. Here, we consider *receiver swaption*.

To do so, we recall that the swap value can be expressed as the difference of fixed rate coupon bond (with coupon rate K and face value N) and floating rate bond (with face value N). That is,

$$V_T^{Swap} = \begin{Bmatrix} Time\ T\ price\ of \\ fixed\ rate\ bond \end{Bmatrix} - \begin{Bmatrix} Time\ T\ price\ of \\ floating\ rate\ bond \end{Bmatrix}.$$

Since the price of the floating rate bond is N, so the terminal payoff of the payer swaption can be expressed as

$$Terminal\ payoff = \max\left(V_T^{Swap}, 0\right) = \underbrace{\max\left(\begin{Bmatrix} Time\ T\ price\ of \\ fixed\ rate\ bond \end{Bmatrix} - N, 0\right)}_{Bond\ call\ option}.$$

It follows from no arbitrage pricing principle that the price of the receiver swaption equals the price of T-year bond call option on coupon bond.

What should we do next?

As we have seen in the previous sections, the prices of callable bond, puttable bond, interest rate cap/floor and swaption can be derived from prices of bond options. To complete the pricing issue, it remains to derive the pricing formula for bond options.

As mentioned before, this requires a mathematical model which can simulate the price movement of the bond price. Since the bond price depends on the market interest rate, thus we need to construct a model that can simulate the movement of the interest rate.

Recently, there are two types of such models:

- Discrete time models (e.g. Binomial tree model and trinomial tree model): This is the simplest pricing model which is easy to use.
- Continuous time models (e.g. Black model and short-rate model): These models can capture the *continuous movement* of the interest rate. However, the pricing requires more advanced mathematics.